nature ecology & evolution

Supplementary information

https://doi.org/10.1038/s41559-025-02873-7

Loss of macroevolutionary species fitness explains the rise and fall of clades

In the format provided by the authors and unedited

Supplementary Text

Constant rate Fossilized Birth-Death model

In the constant rate fossilized birth-death model (CFBD), at any time, all lineages share the same speciation, extinction, and fossilization rate λ, μ, and ψ, respectively (*i.e.*, $\lambda_l(t) = \lambda$, $\mu_l(t) = \mu$ and $\psi_l(t) = \psi$). While there is an analytical solution of this likelihood, here we develop the DA approach, which yields the same exact posterior distribution of the parameters as its closed-form counterpart, but also yields posterior complete trees.

Likelihood The likelihood for a complete unordered crown tree under a CFBD process is simply

$$\ell(\Psi|\lambda,\mu,\psi) = \lambda^{s-1}\mu^{z}\psi^{f}\mathrm{e}^{-(\lambda+\mu+\psi)L},$$

where s, z and f are the number of speciation, extinction and fossilization events, respectively, and L is the tree length (sum of all branches). For a stem tree, we do consider all s speciation events.

Bayesian data augmentation algorithm To integrate over Ψ during MCMC, we sample Ψ_a directly from its conditional distribution given Ψ_o and the model parameters (*i.e.*, Gibbs sampling) using forward simulation. We follow (*1, 2*) to impute probable hidden configurations of unobserved lineages because they went extinct or they were unsampled at the present. This procedure generates *complete trees* (*i.e.*, data augmented trees) which are concordant with the empirical (reconstructed) tree under the diversification model. (*2*) described the proposals for any general diversification model when the empirical tree consisted on extant-only lineages. Here we briefly describe the forward simulation procedure, highlighting the differences when making inference under a fossilized birth-death model.

We iterate over each branch in the empirical tree (a branch constitutes the length of time between speciation or fossilization events) and simulate fossilized birth-death processes forward in time throughout their branch lengths (t_b for branch b), given the current iteration model parameters. For any branch, each initial simulation is a valid proposal if it survives throughout the branch length t_b and if no other

fossilization events occur. For internal branches, if there is more than one surviving lineage by t_b, we uniformly choose one at random and continue the simulation in the other ones. For branches followed by a speciation event the selected lineage represents the one that underwent speciation and is left as is, for branches followed by a fossilization event, the simulation is also pursued on the selected lineage. These simulations remain valid proposals if all unobserved lineages do not leave any fossilization events and either go extinct or are sampled with probability $1 - \rho_b$, where ρ_b is the branch specific sampling fraction (*1, 2*). By default we assume that tip fossils did not leave unsampled descendants at the present.

Let ψ_b denote the forward simulated tree in branch b, $n_b(t_b)$ denote the number of lineages alive at time t_b (*i.e.*, at the end of the branch) and $n_b(0)$ denote the number of lineages alive at the present for ψ_b, then the acceptance ratio in the Metropolis-Hastings (MH) step for the proposed forward simulation ψ_b' if b is a branch of an extant lineage is

$$a = \min\left\{1, \frac{\ell(\Psi'|\lambda,\mu,\psi)\,\ell(\psi_b|\lambda,\mu,\psi)}{\ell(\Psi|\lambda,\mu,\psi)\,\ell(\psi_b'|\lambda,\mu,\psi)} \times \frac{n_b'(0)\rho_b(1-\rho_b)^{n_b'(0)-1}}{n_b(0)\rho_b(1-\rho_b)^{n_b(0)-1}}\right\},$$

and for the other branches b is

$$a = \min\left\{1, \frac{\ell(\Psi'|\lambda,\mu,\psi)\,\ell(\psi_b|\lambda,\mu,\psi)}{\ell(\Psi|\lambda,\mu,\psi)\,\ell(\psi_b'|\lambda,\mu,\psi)} \times \frac{1/n_b(t_b)}{1/n_b'(t_b)} \times \frac{(1-\rho_b)^{n_b'(0)}}{(1-\rho_b)^{n_b(0)}}\right\},$$

where the $1/n_b(t_b)$ factor comes from the proposal density of randomly and uniformly choosing one of the extant lineages at time t_b as the one sampled in the reconstructed tree. These acceptance ratios simplify then for all branches to

$$a = \min\left\{1, \frac{n_b'(t_b)}{n_b(t_b)}(1-\rho_b)^{n_b'(0)-n_b(0)}\right\}.$$

The MH acceptance probability for the sampling fraction is common to all further models, so, for simplicity, we do not show it in the rest of the manuscript.

The model parameters of the process, λ, μ and ψ, are updated using Gibbs sampling using the Gamma conjugate prior, $\Gamma(\kappa, \varsigma)$, for a Poisson process, from which we sample directly

$$p(\lambda|\cdot) = \Gamma(\kappa + s, \varsigma + L),$$

and similarly with μ and ψ but using the number of extinction z and fossilization f events, respectively, in the complete tree of the current iteration. If there is conditioning on survival (see Methods), an additional MH step is needed using the pseudo-marginal approach described in (*2*).

Validation for the CFBD We validate our data augmentation implementation of the constant rate fossilized birth-death (CFBD) by comparing the rate posteriors with the analytical solution for the likelihood from (*3*) implemented in a Bayesian framework in the 'RevBayes' v1.3.0 software (*4*) using Gamma priors $\Gamma(1.0, 1.0)$ across all three parameters (fig. S1).

We also provide a rough comparison of computational efficiency between our DA algorithm in Tapestree and the analytical solution in RevBayes using five empirical trees (one randomly chosen for Acanthuridae, penguins, Siganidae, Scarini and Mauritiinae). For a quick comparison, we evaluated only the Constant FBD model twice on each tree . We used Tapestree.jl version 0.4.1 on Julia v1.11.5 and RevBayes v1.3.0 in an Apple M1 Max (64 GB RAM) and used the default updates (modifying the proportion of each type of updates could lead to different performance). The results, in terms of number of Effective Samples Size (ESS) per second for the parameter with the lowest value is shown in fig. S1. The analytical solution for the constant FBD process is simple and thus remains faster that our DA implementation (although the latter has the advantage of returning posterior *complete* trees, from which we can derive other biological properties, like the diversity through time). The efficiency of DA depends on multiple factors, including the number of DA (unobserved) lineages (higher extinction and lower sampling fraction translates to higher number of DA lineages, and thus of simulations) as well as the survival probability (which is estimated using simulations using the pseudomarginal algorithm described in (*2*)). Data augmentation becomes very useful once the closed solution is costly, impractical or is not known.

The episodic Fossilized Birth-Death model (eFBD)

We expand the CFBD to enable variation of fossilization rates at *a priori* defined periods, which can reflect differences across stratigraphic ranges (*5, 6*), which we call the episodic Fossilized Birth-Death

model (eFBD). While the eFBD can easily accommodate different rates of speciation and extinction during different periods, we simply focus here on relaxing the constancy of fossilization rates as a build-up validation step towards the FBDD process (Methods). Thus, we allow the fossilization rate to vary temporarily following a piecewise constant function across K predefined periods and estimate different fossilization rates delimited by $|k| - 1$ times, t^{ψ}, such that $t_k^{\psi} \in [T_{\Psi}, T_0]$ where rates become $\psi_l(t) = \psi_k$, for $t_k^{\psi} < t < t_{k+1}^{\psi}$. The likelihood for a complete tree under this model then becomes

$$\ell(\Psi|\lambda, \mu, \psi) = \lambda^{s-1}\mu^{z}\mathrm{e}^{-(\lambda+\mu)L}\prod_{k\in K}\psi_k^{f_k}\mathrm{e}^{-\psi_k L_k},$$

where f_k and L_k are the number of fossilization events and tree length, respectively, during period k. The data augmented algorithm is then refined to accommodate this new source of variation by taking into account the change in rates when traversing periods.

Simulation study for the eFBD We explored the statistical behavior and validity of our inference algorithm under the eFBD using simulations. We simulated trees for a duration of 6 time units with three periods of equal duration ($K = 3$), from $[6, 4]$, $[4, 2]$ and $[2, 0]$ under 4 simulation scenarios with different temporal variation of fossilization rates: *i)* rates remain constant (*i.e.*, $\psi_1 = \psi_2 = \psi_3 = 0.2$), *ii)* rates increase (*i.e.*, $\psi_1 = 0.05; \psi_2 = 0.2; \psi_3 = 0.5$), *iii)* rates decrease and then increase (*i.e.*, $\psi_1 = 0.5; \psi_2 = 0.1; \psi_3 = 0.5$), and *iv)* rates decrease (*i.e.*, $\psi_1 = 0.5; \psi_2 = 0.2; \psi_3 = 0.05$). For each scenario we simulated 50 trees of size between 200 and 500 tips (either alive or fossil), and set $\lambda = 1.0$ and $\mu = 0.5$. These values were chosen for biological realism, based on typical values estimated from our empirical analyses. We then made inference with Gamma priors $\Gamma(1.0, 1.0)$ across all parameters using our DA algorithm by running 5.2×10^5 iterations, discarding the first 2×10^4 as burnin and sampling every 5×10^2 iterations. The results are shown in fig. S2. Overall, the inference under the eFBD model shows good statistical behavior, with good accuracy (Mean Absolute Error, MAE, being below 0.15 across parameters) and over 90% coverage for λ and ψ, and over 86% coverage for μ across all simulations. As expected the variance for ψ_k increases as one moves further back in time (*i.e.*,

variance(ψ_1) > variance(ψ_2) > variance(ψ_3); fig. S2).

Metropolis-Hasting (MH) ratios for the FBDD model

Here we describe those additional MH ratios for the FBDD on top of those already described for the BDD model in (*2*). Let $\Psi, \Lambda, \mathrm{M}$ represent the topology, the latent speciation GBM and the latent extinction GBM, respectively, then the simplified acceptance ratio for a valid forward simulation proposal for a branch pr that ends in a sampled ancestor with subsequent branch d is

$$a = \min\left\{1, \frac{p(\lambda(t_{pr})|\lambda(t_d), \alpha_\lambda\sigma_\lambda)\, p(\mu(t_{pr})|\mu(t_d), \alpha_\mu\sigma_\mu)}{p(\lambda(t_{pr})'|\lambda(t_d), \alpha_\lambda\sigma_\lambda)\, p(\mu(t_{pr})'|\mu(t_d), \alpha_\mu, \sigma_\mu)} \times \frac{\ell(\Lambda', \mathrm{M}'|\Psi, \alpha_\lambda, \alpha_\mu, \sigma_\lambda, \sigma\mu, \psi, \mathcal{M}_{\text{FBDD}})}{\ell(\Lambda, \mathrm{M}|\Psi, \alpha_\lambda, \alpha_\mu, \sigma_\lambda, \sigma\mu, \psi, \mathcal{M}_{\text{FBDD}})}\right\},$$

where

$$p(\lambda(t_{pr})|\lambda(t_d), \alpha_\lambda, \sigma_\lambda) = \lambda(t_{pr}) \sim \mathrm{N}\left(\lambda(t_d) - \alpha_\lambda t_d, t_d\sigma_\lambda^2\right),$$

and similarly for $p(\mu(t_{pr})|\mu(t_d), \alpha_\mu, \sigma_\mu)$. For a tip fossil, the forward simulation is accepted or rejected based on it being a valid proposal as described in the Methods.

Simulation study for the FBDD

We conducted a simulation study to explore the behavior of the FBDD under our DA inference algorithm. Following (*2*), we take advantage of the following approximation to simulate under the FBDD: let V be a random variable for the time of an event and let r_i and r_{i+1}, be the event rate at time t_i and t_{i+1}, respectively, where $t_{i+1} - t_i = \delta \geq 0$, then we have

$$p(t_i \leq V < t_{i+1} \mid t_i \leq V) \approx \bar{r}_{i,i+1}\delta.$$

We first conducted simulations with constant fossilization rates. To simultaneously explore the role of increasing fossil sampling across parameter space, we considered 5 scenarios with incremental fossilization rate. Specifically, we set ψ to be $\{0.05, 0.1, 0.5, 1.0, 2.0\}$ and, for each fossil rate scenario, simulated 100 FBDD trees of 150 extant species by sampling from a range of parameter space (ED fig.

3), yielding a total of 500 simulations. For each simulation, we set the initial rates of speciation $\lambda_0 = 0.3$ and extinction $\mu_0 = 0.25$, which then vary according to the GBM. For the parameters of the GBM process, we sample α_λ and α_μ from the uniform distribution $\mathrm{U}(-0.25, 0.25)$ and σ_λ and σ_μ from the uniform distribution $\mathrm{U}(0.1, 0.6)$. As above, these values were chosen for biological realism, based on typical values estimated from our empirical analyses. We start each simulation with two lineages and make sure that at least one survived to the present. We then sample a tree with a determined number of extant species following (*7*) to avoid biases.

We then conducted simulations with varying fossilization rates. Specifically, we simulated over 9 time units where fossilization rates, $\psi(t)$, changed at three periods, from $[9, 6)$, $[6, 3)$ and $[3, 0]$, 0 being the present, and retained those that ended up with a number of tips between 150 and 350. For these epochs we used 3 scenarios where fossilization rates varied differently: i) Increasing fossilization rates with time where $\psi(t) = \{0.2, 0.5, 1.0\}$ for each period, respectively; ii) Decrease and increase of fossilization rates with time where $\psi(t) = \{1.0, 0.5, 1.0\}$ for each period, respectively; and iii) Decreasing fossilization rates with time where $\psi(t) = \{1.0, 0.5, 0.2\}$ for each period, respectively. Here we had to expand the parameter combination space so that, during the specified time we can sample trees with the specified number of tips (*i.e.*, $[150, 350]$). We sampled α_λ and α_μ from the uniform distribution $\mathrm{U}(-0.2, 0.2)$, λ_0 from the uniform distribution $\mathrm{U}(0.05, 1)$, μ_0 from $\mathrm{U}(0.5, 0.8) \times \lambda_0$ (so extinction being a proportion between 0.5 and 0.8 of the speciation at the root, and σ_λ and σ_μ from the uniform distribution $\mathrm{U}(0.1, 0.4)$. For a given combination of parameters we tried 10 times to obtain a successful simulations, then we resampled the governing parameters.

For each simulation we then conducted inference specifying weakly informative priors for most parameters. Specifically, we used log-normal prior with very large variance for λ_0 and μ_0 (*i.e.*, effectively flat), an Inverse Gamma, $\Gamma^{-1}(0.05, 0.05)$, for σ_λ,$\Gamma^{-1}(3, 0.1)$ for σ_μ, and a Gaussian $\mathrm{N}(0, 1)$ for α_λ and α_μ. We note that a weakly informative prior on σ_μ helps avoid pathological behavior when, for low values of ψ and by chance, the resulting empirical tree has few tip fossils. Note that the number of fossil

tips is very important for extinction rates, since a tip fossil forces at least one extinction event (effectively removing any probability density on extinction rates being 0). In empirical trees, the ratio of tip fossils to sampled ancestors is usually higher than expected from FBD simulations, because researchers usually include only one fossil occurrence per extinct species (excluding multiple occurrences, if available, and thus under-representing the amount of sampled ancestors). While fossils in general yield better extinction estimates, an under-representation of sampled ancestors in empirical trees conflicts with the assumptions from the FBD process, as explained in the main Methods and below. We run each simulation for 3.5×10^6 iterations, discarding the first 5×10^5 as burnin and sampling every 3×10^3 iterations.

Statistical coverage and accuracy Overall, we find good statistical behavior from our inference algorithm of the FBDD model (ED fig. 3-4). First, as expected from our simulation from eCFB, we can reliably retrieve the fossilization rates ψ_k. Second, and as expected from results from extant-only trees (*2*), α_λ, α_μ, σ_λ and Λ (the whole speciation rates diffusion) show good coverage and accuracy across all fossilization scenarios (ED fig. 3-4). We find that M, the extinction rates diffusion, performs much better as the fossilization rate increases. Specifically, at $\psi = 0.05$ and $\psi = 0.1$ there is an underestimation of extinction rates, and while median coverage is around 95%, many simulations have a much lower coverage (ED fig. 2). However, with fossilization rates $\psi > 0.5$, or with varying fossilization rates where the minimum was 0.2 but some periods experienced (0.5 or 1.0), the statistical behavior of M increases sharply, with almost no bias, small relative error and good coverage (ED fig. 3-4). Finally, while the accuracy of σ_μ is very low, coverage remains moderate (over 50%), implying that while extinction is retrievable, its rate of diffusion is not, and the posterior distribution integrates over much of its parameter space (ED fig. 3-4).

References

1. O. Maliet, H. Morlon, *Systematic biology* **71**, 353 (2022).

2. I. Quintero, N. Lartillot, H. Morlon, *Science* **384**, 1007 (2024).

3. T. Stadler, *Journal of theoretical biology* **267**, 396 (2010).

4. S. Höhna, *et al.*, *Systematic biology* **65**, 726 (2016).

5. S. M. Kidwell, S. M. Holland, *Annual Review of Ecology and Systematics* **33**, 561 (2002).

6. D. Silvestro, N. Salamin, J. Schnitzler, *Methods in Ecology and Evolution* **5**, 1126 (2014).

7. T. Stadler, *Systematic biology* **60**, 676 (2011).

8. T. Stadler, A. Gavryushkina, R. C. Warnock, A. J. Drummond, T. A. Heath, *Journal of theoretical biology* **447**, 41 (2018).

9. W. Pett, T. A. Heath, *Phylogenetics in the Genomic Era*, C. Scornavacca, F. Delsuc, N. Galtier, eds. (No commercial publisher | Authors open access book, 2020), pp. 5.1:1–5.1:18.

10. U. Stolz, A. Gavryushkina, T. G. Vaughan, T. Stadler, B. J. Allen, *bioRxiv* (2025).

11. J. Andréoletti, *et al.*, *Systematic Biology* **71**, 1440 (2022).

12. A. Ruebenstahl, N. Mongiardino Koch, J. C. Lamsdell, D. E. Briggs, *Proceedings of the Royal Society B* **291**, 20241184 (2024).

13. C. D. Brownstein, R. C. Harrington, T. J. Near, *Journal of Biogeography* **50**, 1191 (2023).

14. H. R. Grunert, N. Brocklehurst, J. Fröbisch, *Scientific Reports* **9**, 5063 (2019).

15. T. R. Simões, M. W. Caldwell, S. E. Pierce, *BMC biology* **18**, 1 (2020).

16. B. M. Farina, P. L. Godoy, R. B. Benson, M. C. Langer, G. S. Ferreira, *Ecology and Evolution* **13**, e10201 (2023).

17. R. B. Benson, R. A. Frigot, A. Goswami, B. Andres, R. J. Butler, *Nature communications* **5**, 3567 (2014).

18. G. Lloyd, D. Bapst, M. Friedman, K. Davis, *Biology letters* **12**, 20160609 (2016).

19. E. W. Wilberg, A. H. Turner, C. A. Brochu, *Scientific reports* **9**, 514 (2019).

20. Q. Zhang, R. H. Ree, N. Salamin, Y. Xing, D. Silvestro, *Systematic Biology* **71**, 242 (2022).

21. E. M. Troyer, *et al.*, *Proceedings of the National Academy of Sciences* **119**, e2122486119 (2022).

22. C. D. Bacon, *et al.*, *Biology Letters* **18**, 20220214 (2022).

23. A. C. Siqueira, D. R. Bellwood, P. F. Cowman, *Journal of Biogeography* **46**, 1611 (2019).

24. A. Gavryushkina, *et al.*, *Systematic biology* **66**, 57 (2017).

25. J. L. Cantalapiedra, *et al.*, *Nature Ecology & Evolution* **5**, 1266 (2021).

26. O. Sanisidro, M. C. Mihlbachler, J. L. Cantalapiedra, *Science* **380**, 616 (2023).

27. G. T. Lloyd, G. J. Slater, *Systematic Biology* **70**, 922 (2021).

28. S. Hotaling, M. L. Borowiec, L. S. Lins, T. Desvignes, J. L. Kelley, *Molecular phylogenetics and evolution* **162**, 107211 (2021).

29. J. L. Cantalapiedra, *et al.*, *Proceedings of the Royal Society B* **286**, 20182896 (2019).

30. M. Cascini, K. J. Mitchell, A. Cooper, M. J. Phillips, *Systematic Biology* **68**, 520 (2019).

31. L. Varela, P. S. Tambusso, H. G. McDonald, R. A. Fariña, *Systematic Biology* **68**, 204 (2019).

32. T. J. Near, D. Kim, *Molecular Phylogenetics and Evolution* **161**, 107156 (2021).

33. D. Silvestro, *et al.*, *Systematic Biology* **68**, 78 (2018).

193 34. T. Park, *et al.*, *Evolution* p. qpae061 (2024).

194 35. H. A. Ogilvie, *et al.*, *Systematic Biology* **71**, 208 (2022).

195 36. J. L. Cantalapiedra, J. L. Prado, M. Hernández Fernández, M. T. Alberdi, *Science* **355**, 627 (2017).

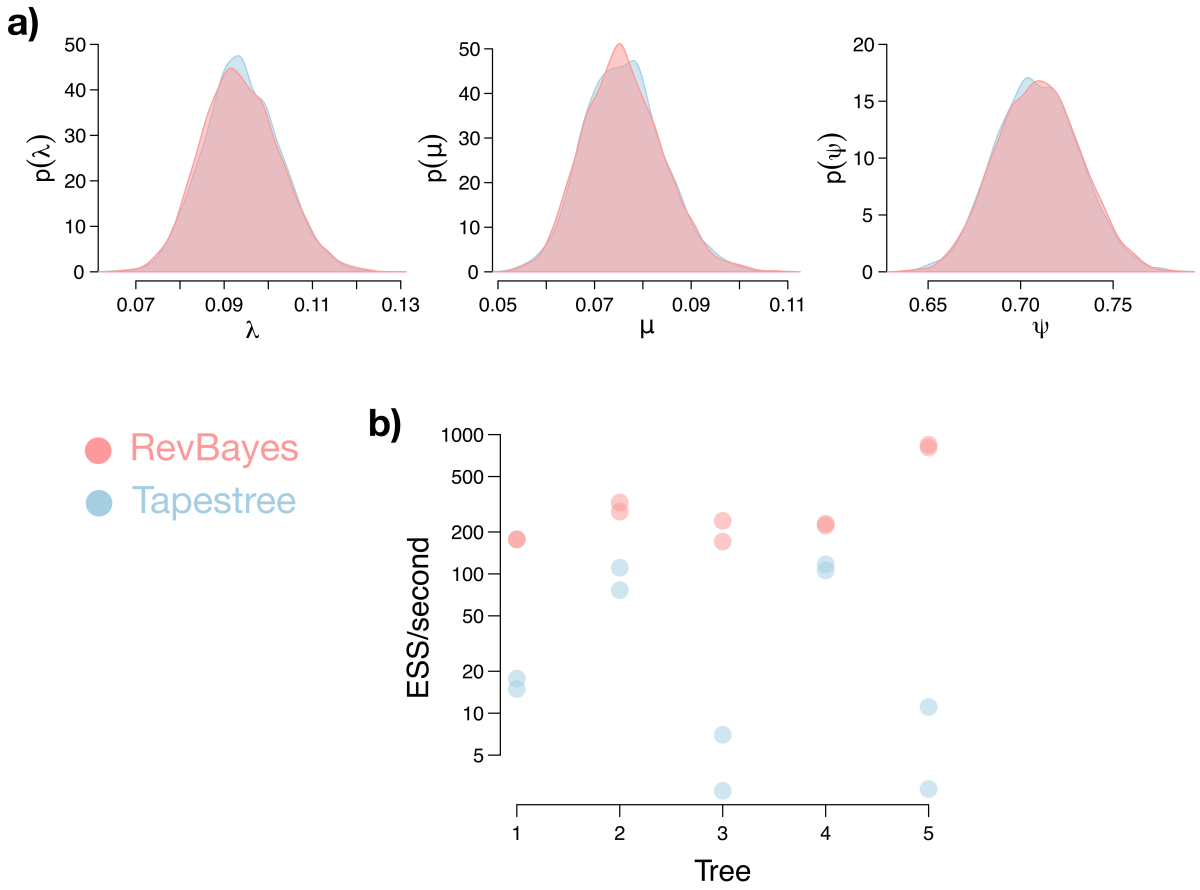
196 37. H. P. Püschel, O. C. Bertrand, J. E. O'reilly, R. Bobe, T. A. Püschel, *Nature ecology & evolution* **5**,
197 808 (2021).

Supplementary Tables

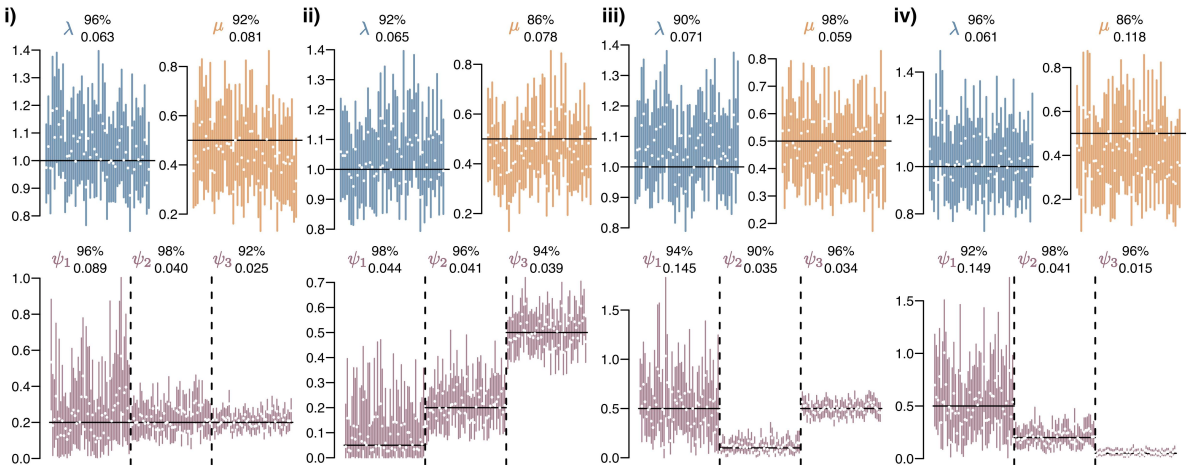
Table S1: Taxonomic groups used in this study with their respective number of extinct and extant species represented, their present-day sampling fraction (ρ) and the reference from where the phylogenetic trees were obtained.

taxon	n extinct	n extant	ρ	tree source
Eurypterida	153	0	1	(*12*)
Dipnoi	16	5	0.87	(*13*)
Therocephalia	51	0	1	(*14*)
Sphenodontidae	34	1	1	(*15*)
Testudinata	448	327	0.49	(*16*)
Pterosauria	109	0	1	(*17*)
Dinosauria (non-avian)	624	0	1*	(*18*)
Crocodylomorpha	127	15	0.55	(*19*)
Juglandaceae	130	47	0.77	(*20*)
Tetraodontiformes	52	187	0.53	(*21*)
Mauritiinae	9	9	0.9	(*22*)
Acanthuridae	25	72	0.9	(*23*)
Siganidae	10	24	0.8	(*23*)
Spheniscidae	36	19	0.95	(*24*)
Proboscidea	182	3	1	(*25*)
Brontotheriidae	57	0	1	(*26*)
Cetacea	405	89	0.97	(*27*)
Zoarcoidei	9	222	0.51	(*28*)
Ruminantia	1064	213	1	(*29*)
Sthenurinae	6	28	1	(*30*)
Folivora	62	2	0.28	(*31*)
Centrarchidae	19	40	0.98	(*32*)
Platyrrhini	33	88	0.44	(*33*)
Pinnipedia	89	36	1	(*34*)
Caninae	42	36	1	(*35*)
Equinae	130	8	1	(*36*)
Hominini	22	1	1	(*37*)

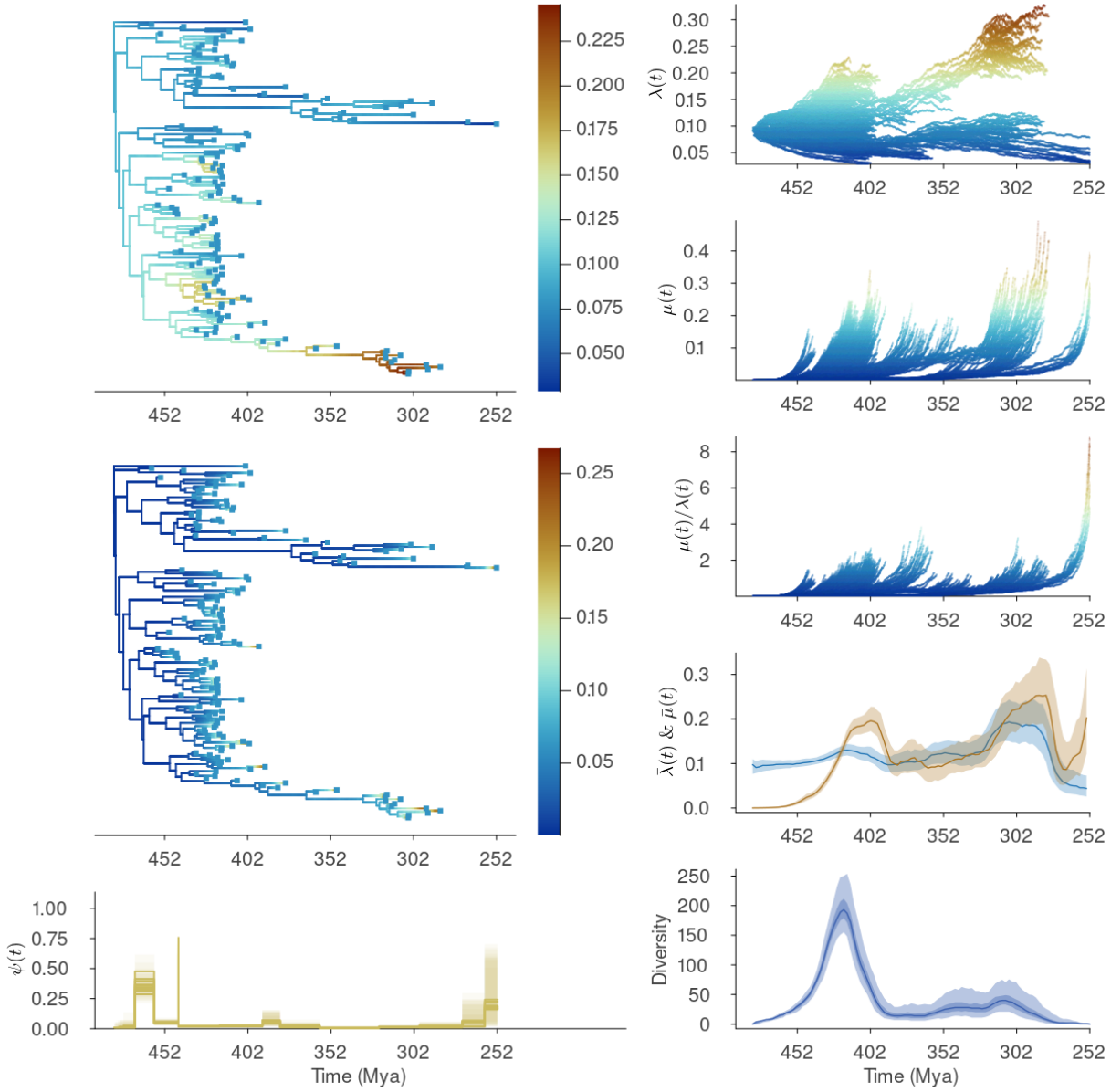
* We analyze non-avian dinosaurs as a clade, acknowledging that we excluded Aves. In this context, all the diversity of non-avian dinosaurs is sampled (thus $\rho = 1$).



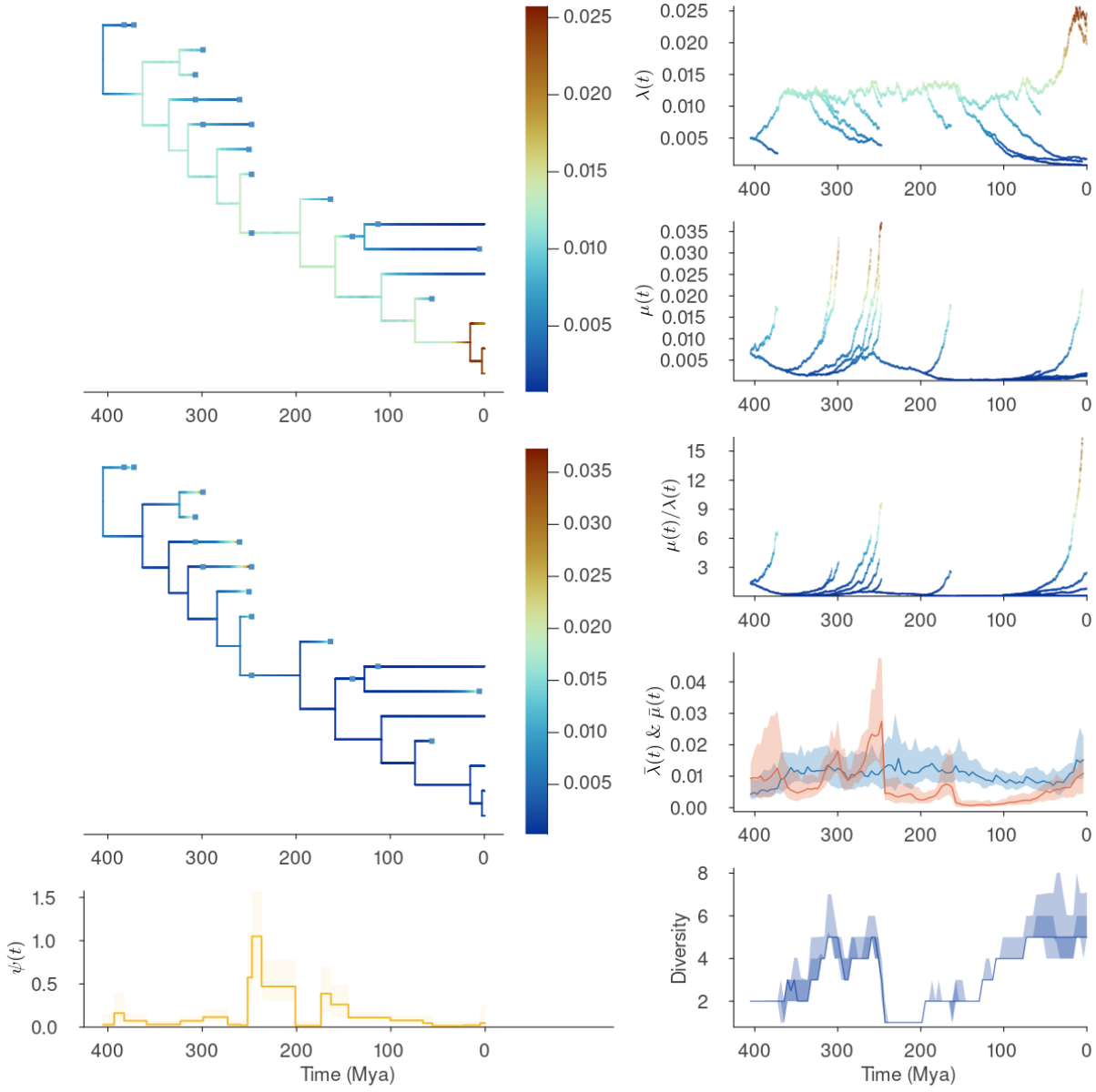
Supplementary Figure 1: Validation of Bayesian data augmentation inference of the fossilized birth-death (CFBD) For the same Posterior distributions for: *left* speciation rate (λ), *middle* extinction rate (μ) and *right* fossilization rate (ψ) from the analytical solution of the likelihood using RevBayes (*4*) in red and from the Bayesian data augmentation algorithm developed in this study in blue.



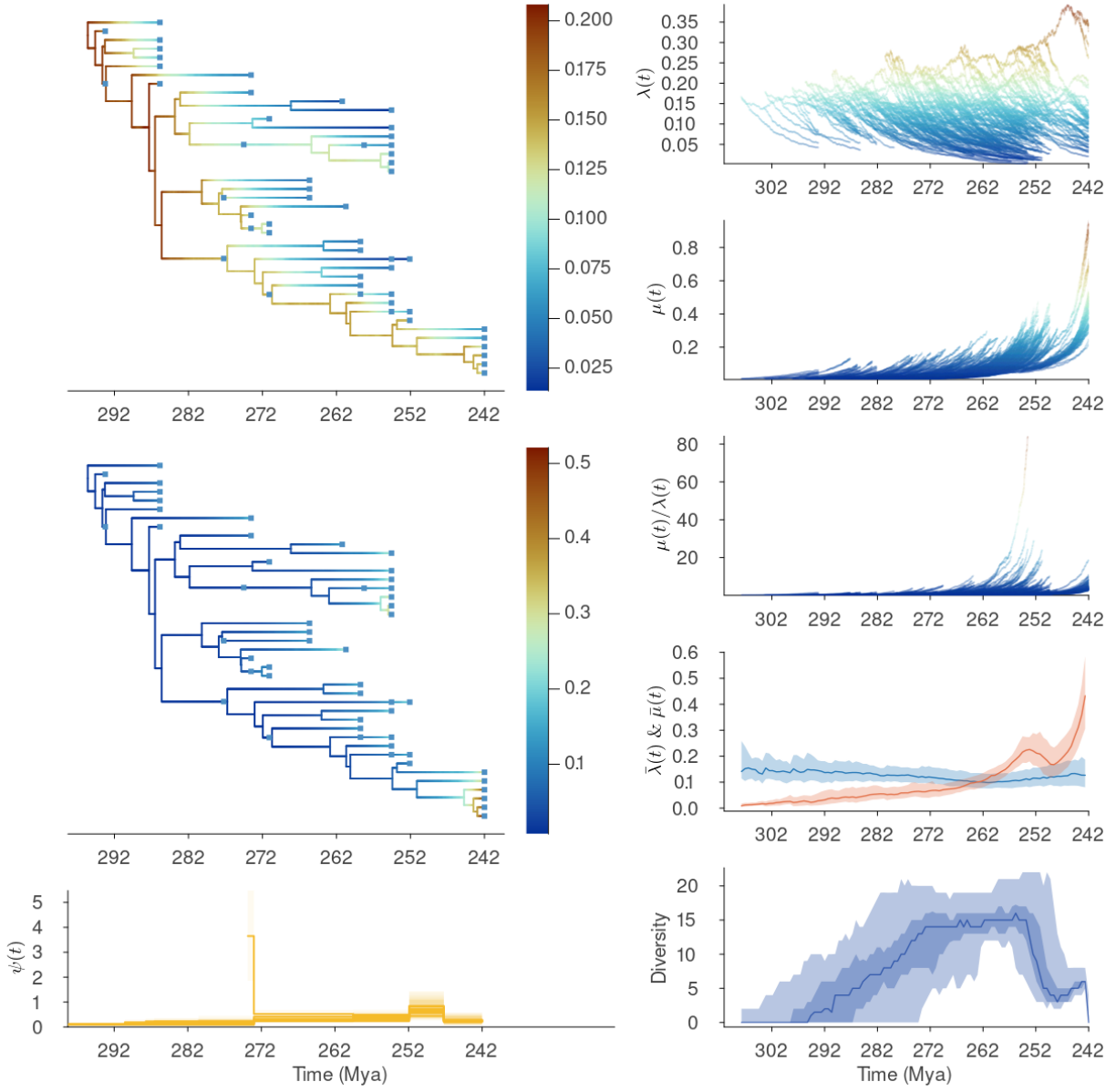
Supplementary Figure 2: Validation of episodic fossilized birth-death (eFBD) Results for each of the 4 scenarios (*i-iv*, see Supplementary Text) in the simulation study for inference under the eFBD. For each scenario, the top row shows speciation and extinction results, while the bottom row shows the results for fossilization rates across three periods. For each parameter, the black horizontal bar shows the true value, and for each of the 50 replicates vertical colored bars show the 95% Credible Interval (CI) and the white point the mean. The top number percentage represents the coverage (proportion of simulations in which the true value was within the CI) and the bottom number the Mean Absolute Error (MAE).



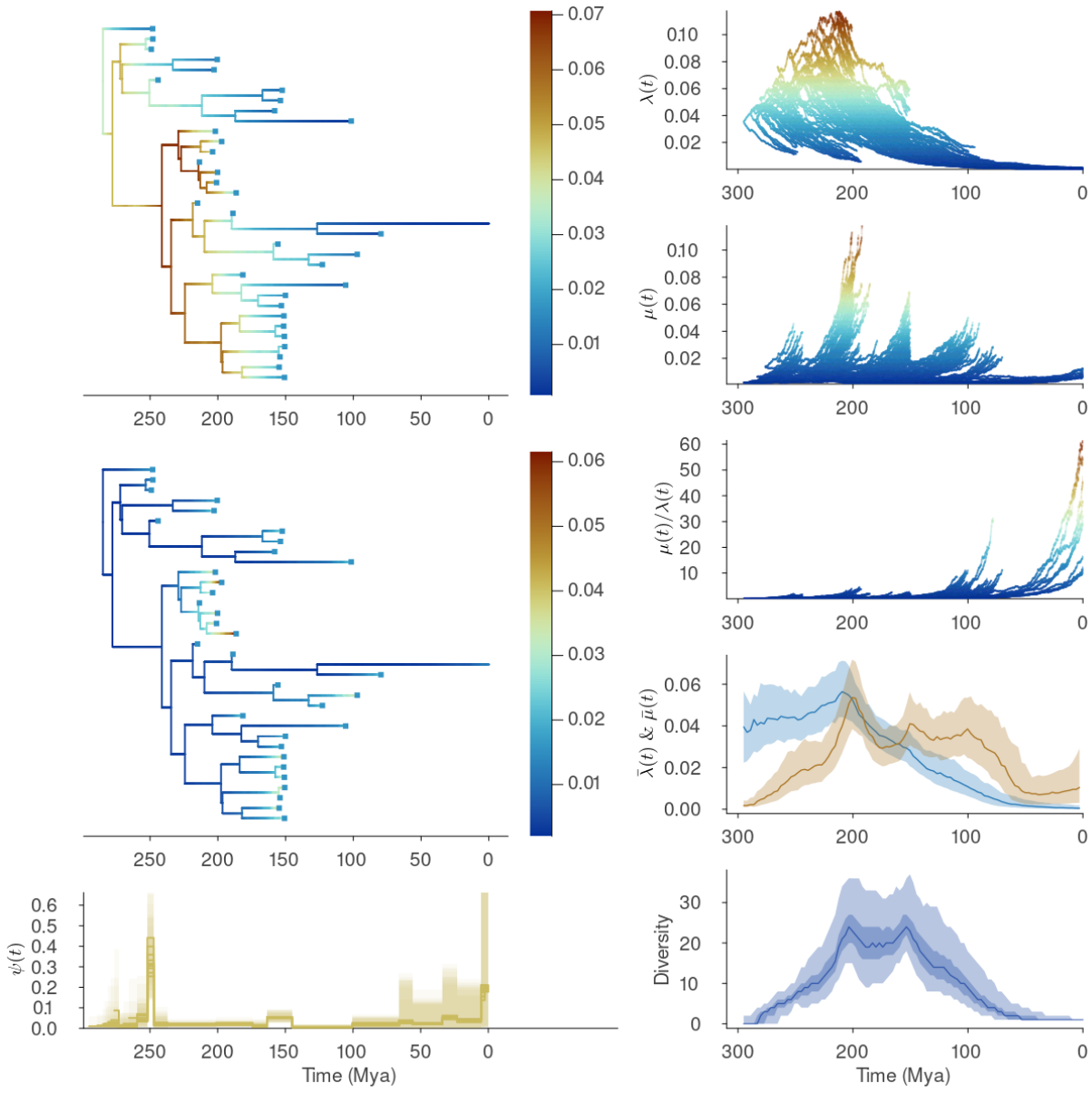
Supplementary Figure 3: FBDD results for Eurypterida. *Left* column shows: *top*, the posterior average rates for speciation and *middle* for extinction rates on the reconstructed tree of one of the empirical tree samples (Note that marginal posterior distributions for rates can only be estimated for the observed (reconstructed) part of the tree (*i.e.*, Ψ_o)). *Bottom* shows the posterior fossilization rates estimates through time for each of the empirical tree samples. *Right* column from *top* to *bottom* shows: marginal posterior median lineage rates of speciation, of extinction and of turnover ($\mu(t)/\lambda(t)$) across the empirical tree samples (again, these three can only be estimated on the observed part of the tree), posterior cross-lineage average speciation ($\bar{\lambda}(t)$ in blue) and extinction ($\bar{\mu}(t)$ in red) rates across the complete trees (average and 95% Credible Interval), and, finally, the median, first quartile and 95% CI of estimated diversity trajectories.



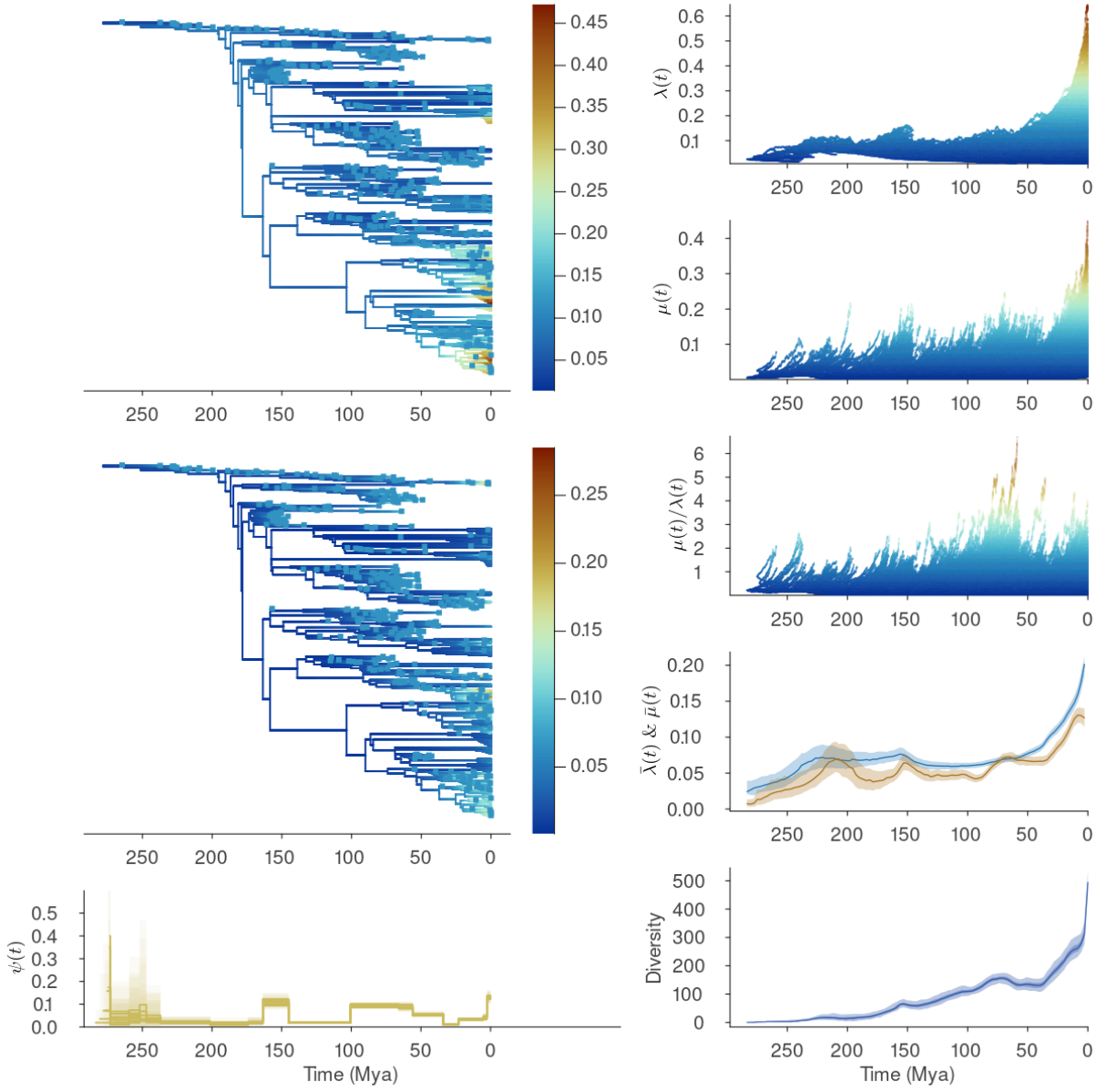
Supplementary Figure 4: FBDD results for Dipnoi. Details as in fig. S3.



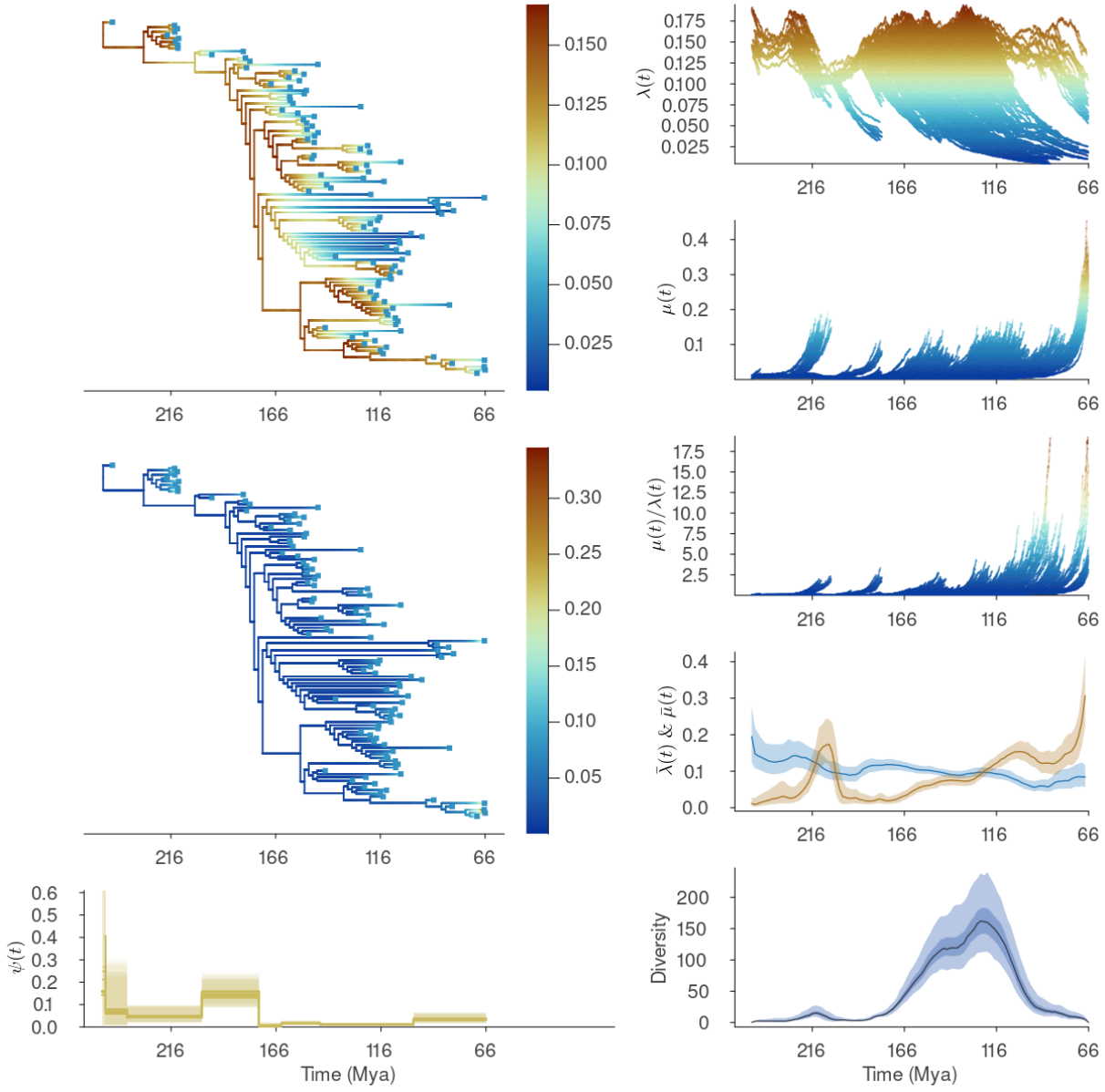
Supplementary Figure 5: FBDD results for Therocephalia. Details as in fig. S3.



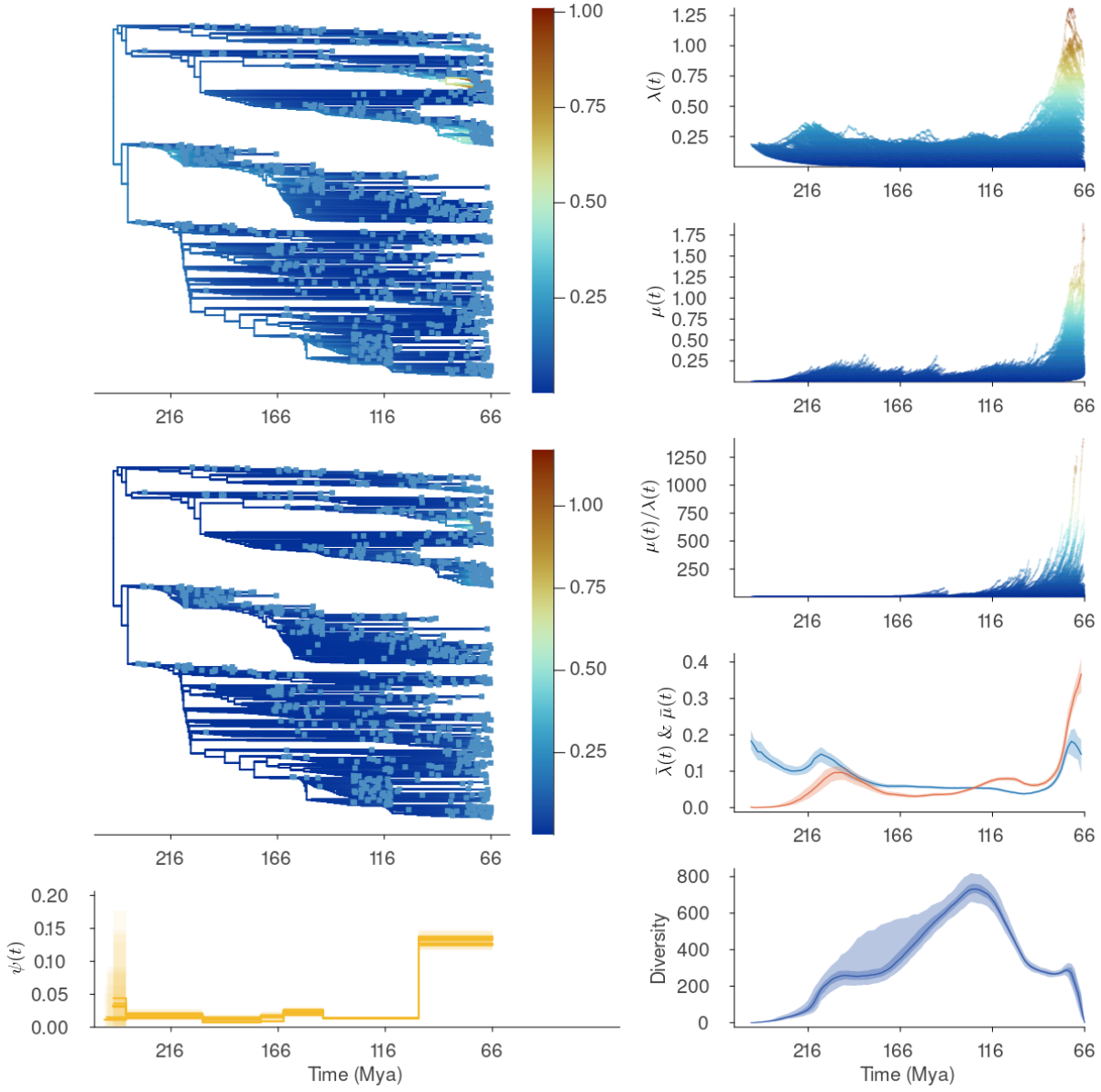
Supplementary Figure 6: FBDD results for Sphenodontidae. Details as in fig. S3.



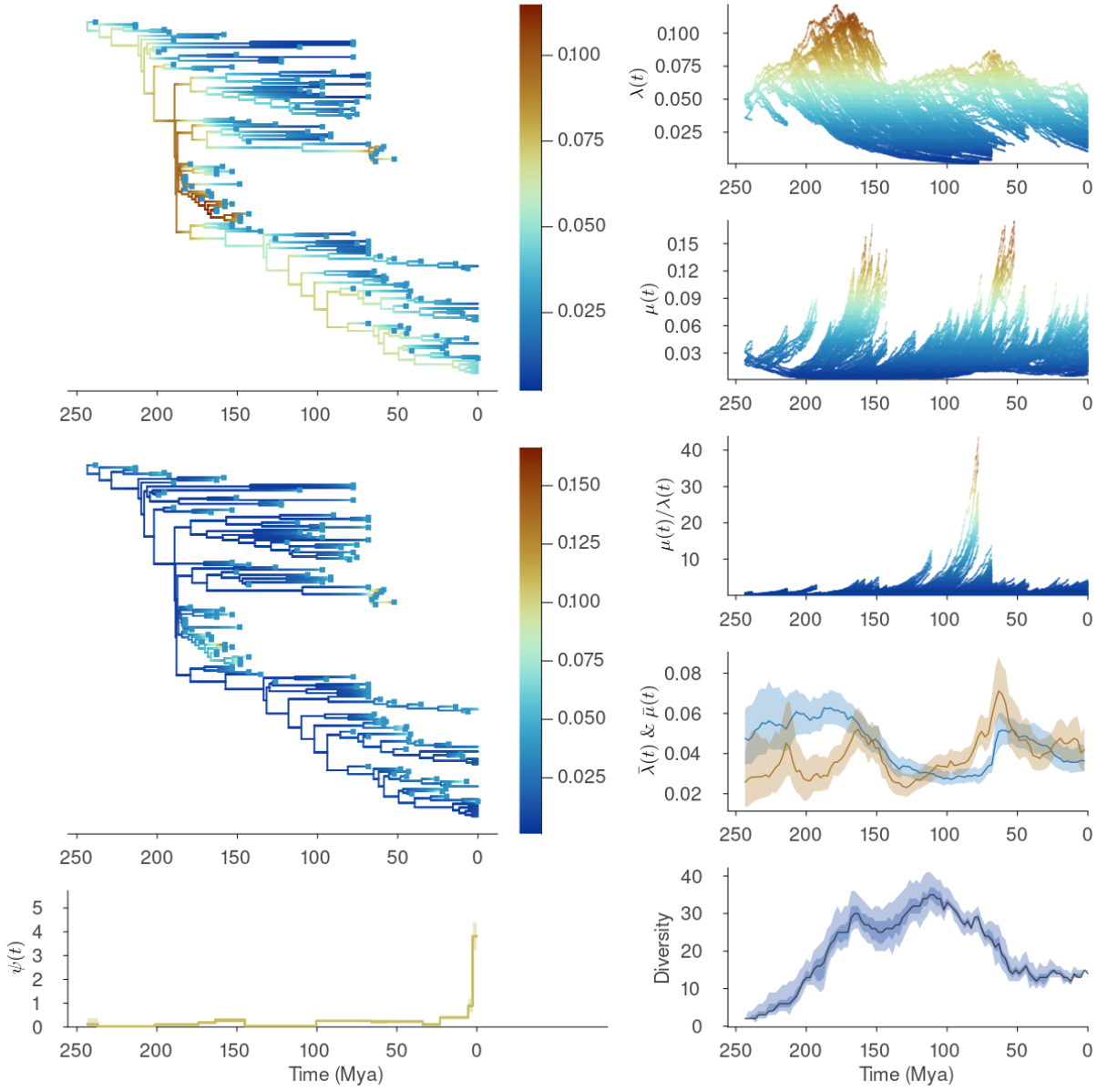
Supplementary Figure 7: FBDD results for Testudinata. Details as in fig. S3.



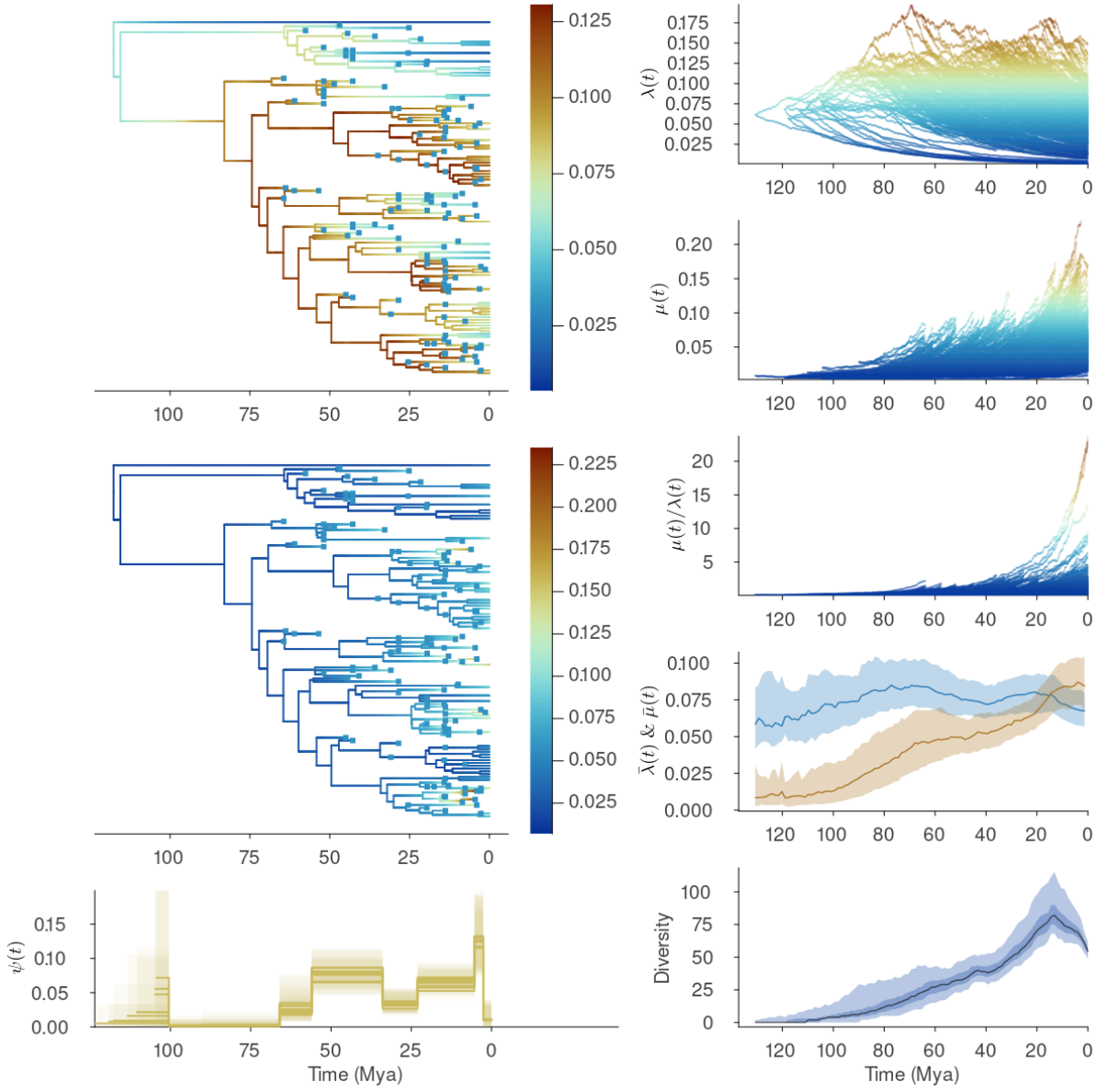
Supplementary Figure 8: FBDD results for Pterosauria. Details as in fig. S3.



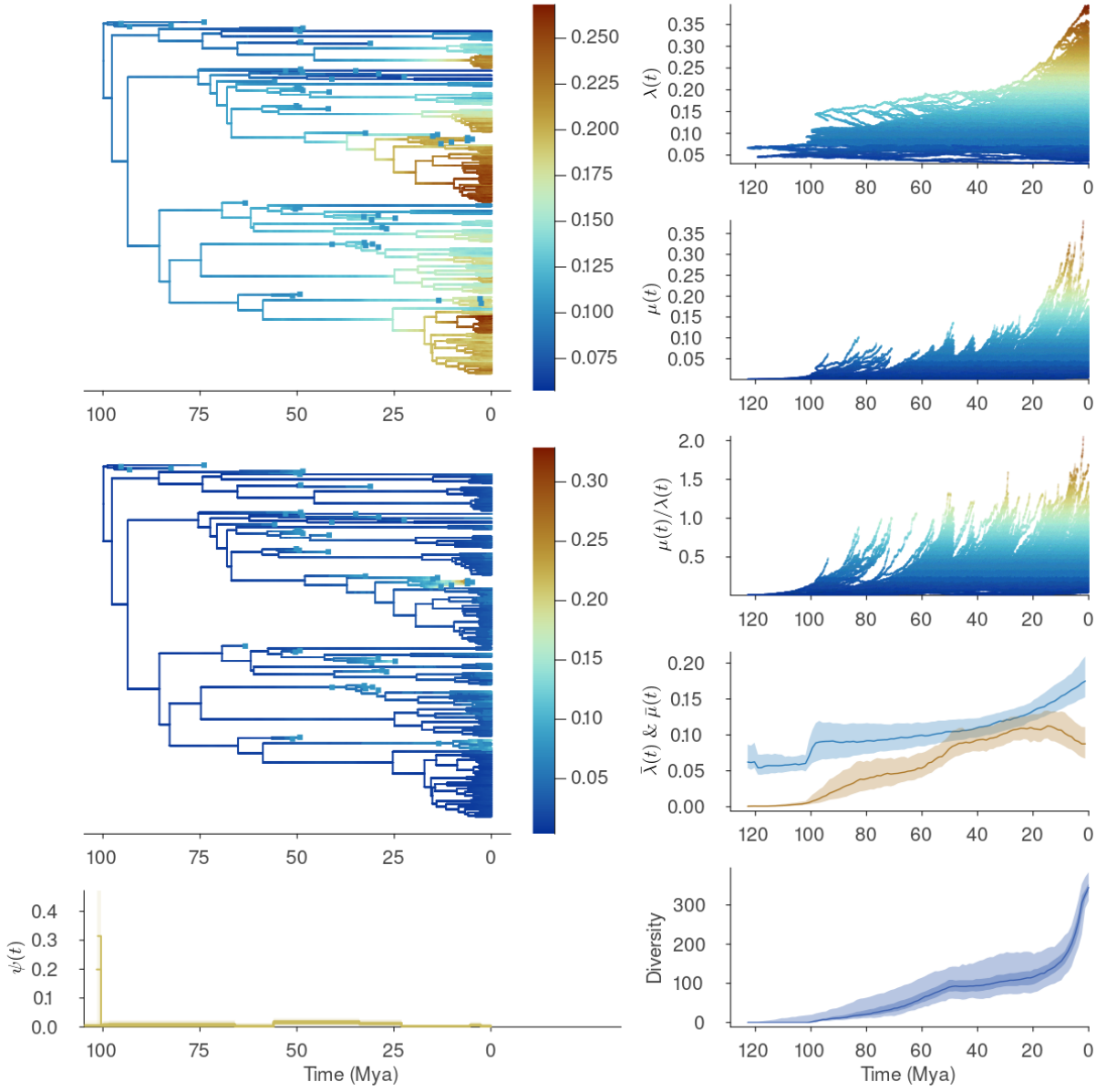
Supplementary Figure 9: FBDD results for Dinosauria. Details as in fig. S3.



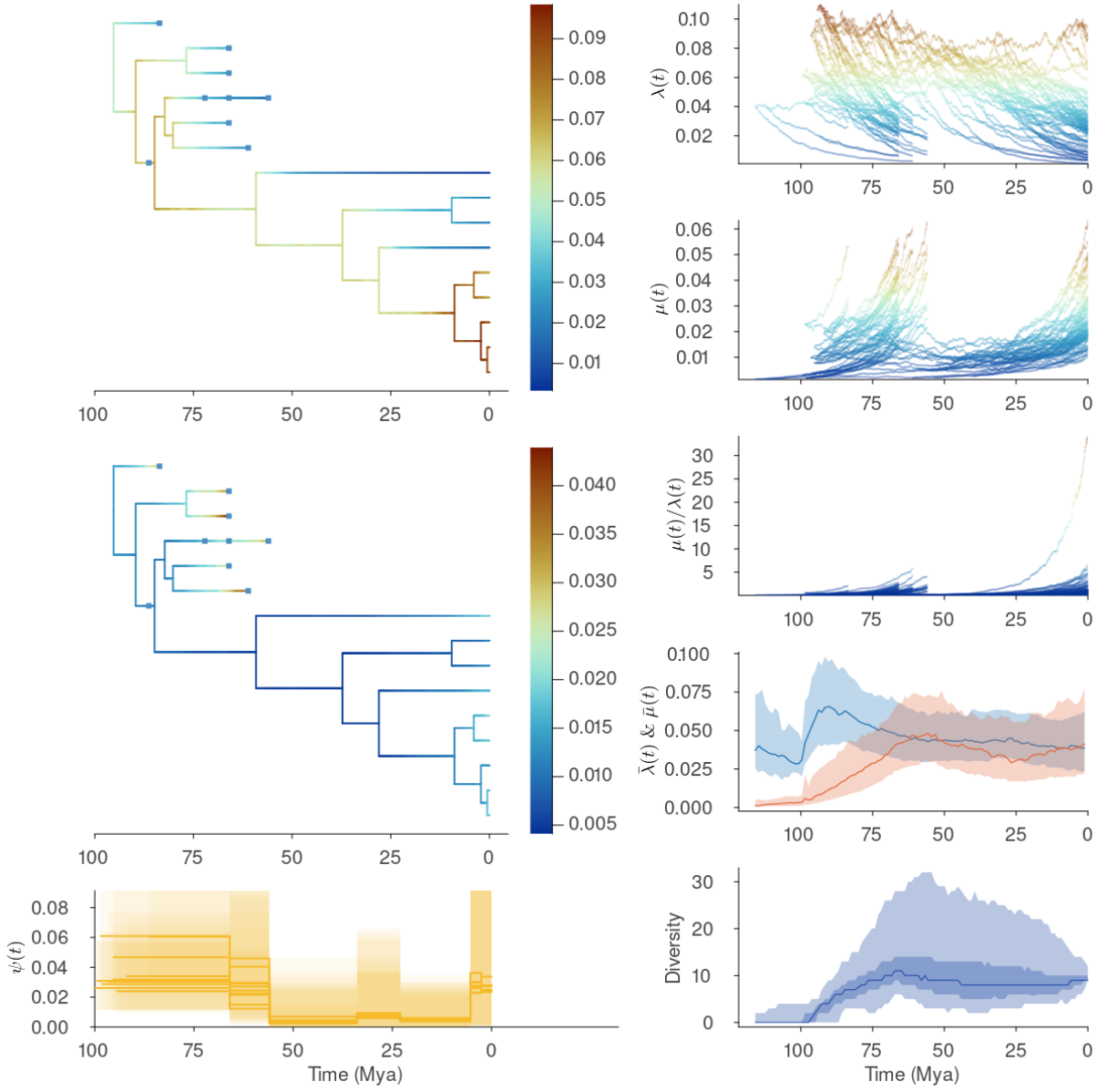
Supplementary Figure 10: FBDD results for Crocodylomorpha. Details as in fig. S3.



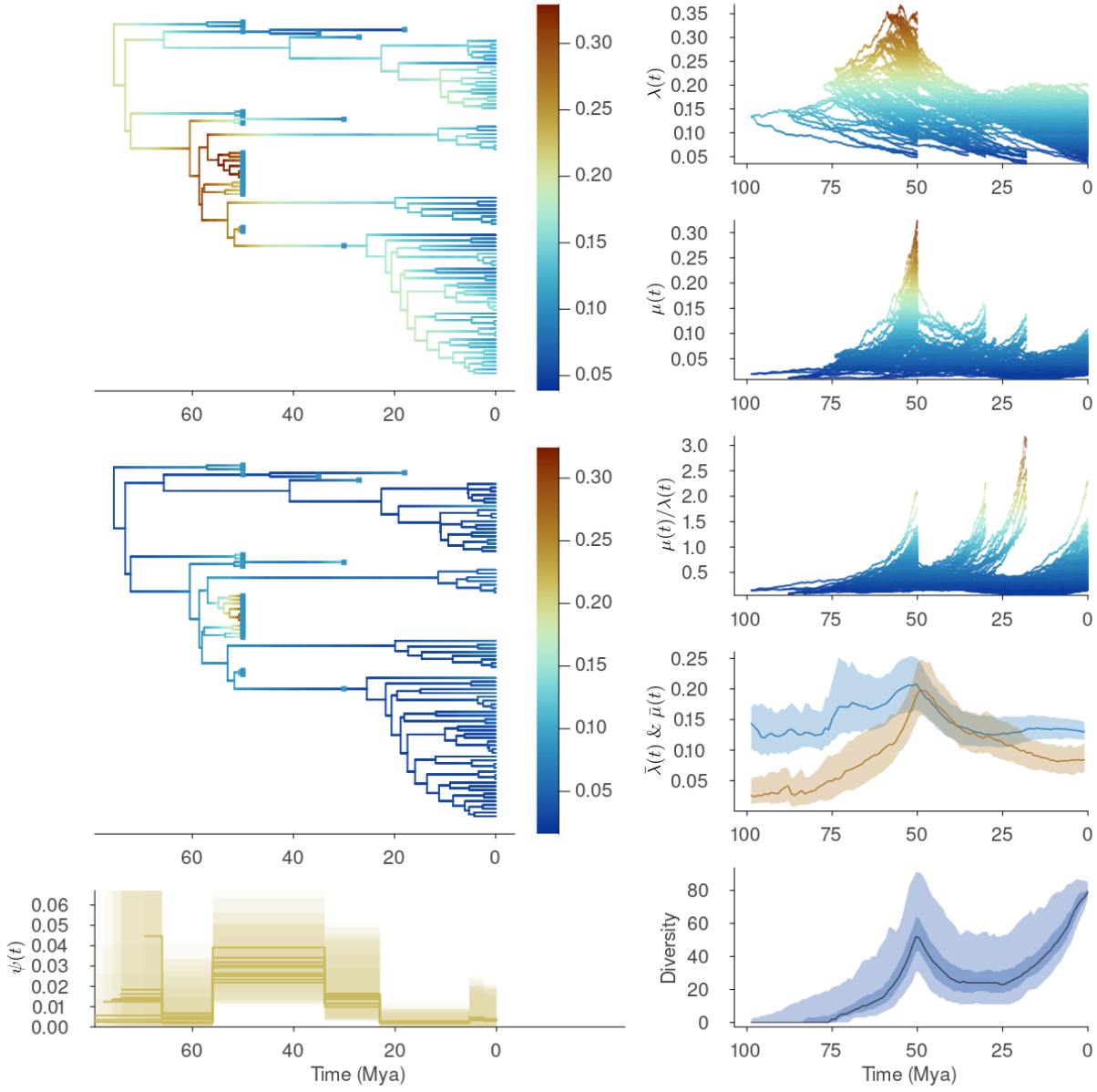
Supplementary Figure 11: FBDD results for Juglandaceae. Details as in fig. S3.



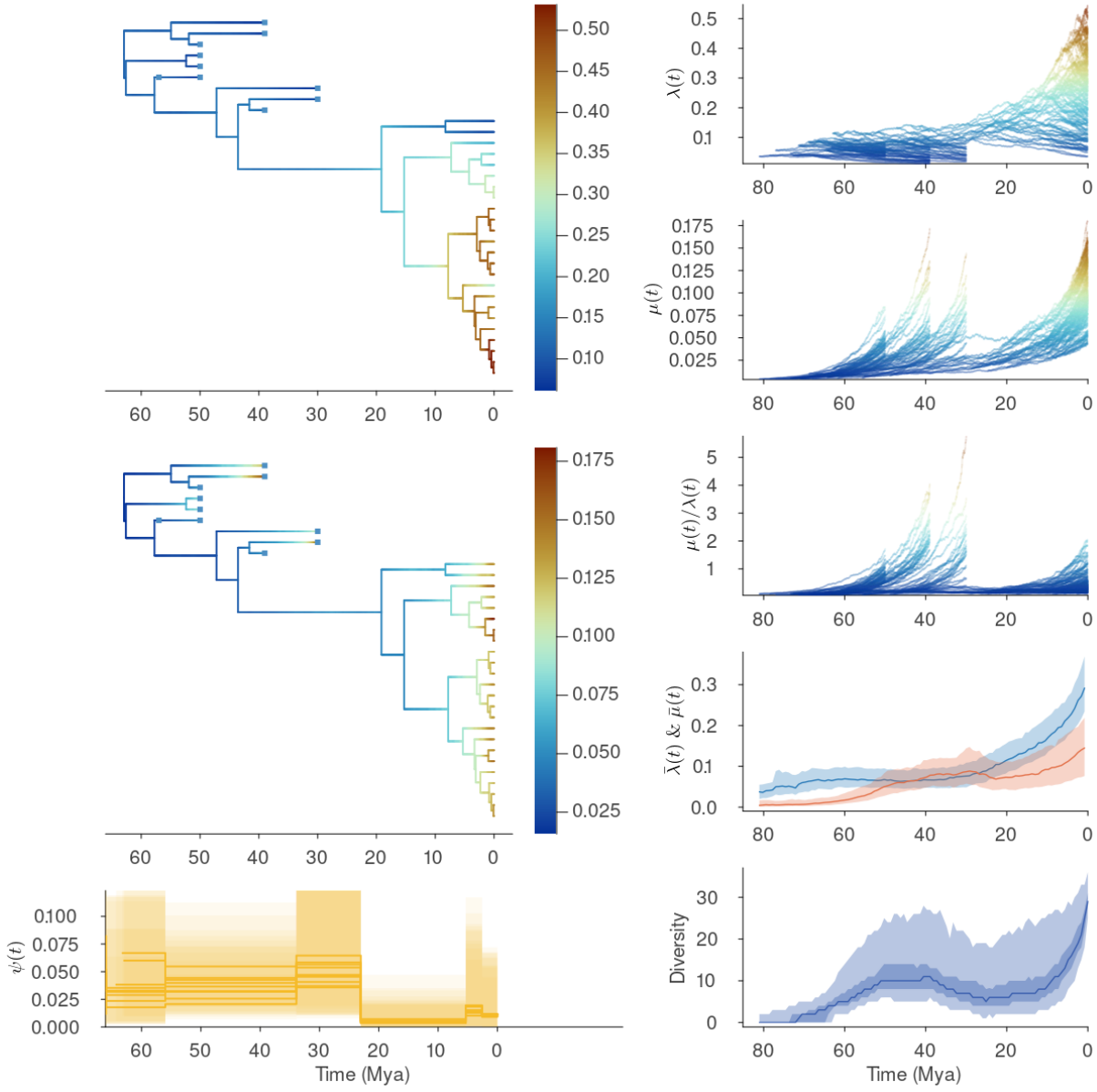
Supplementary Figure 12: FBDD results for Tetraodontiformes. Details as in fig. S3.



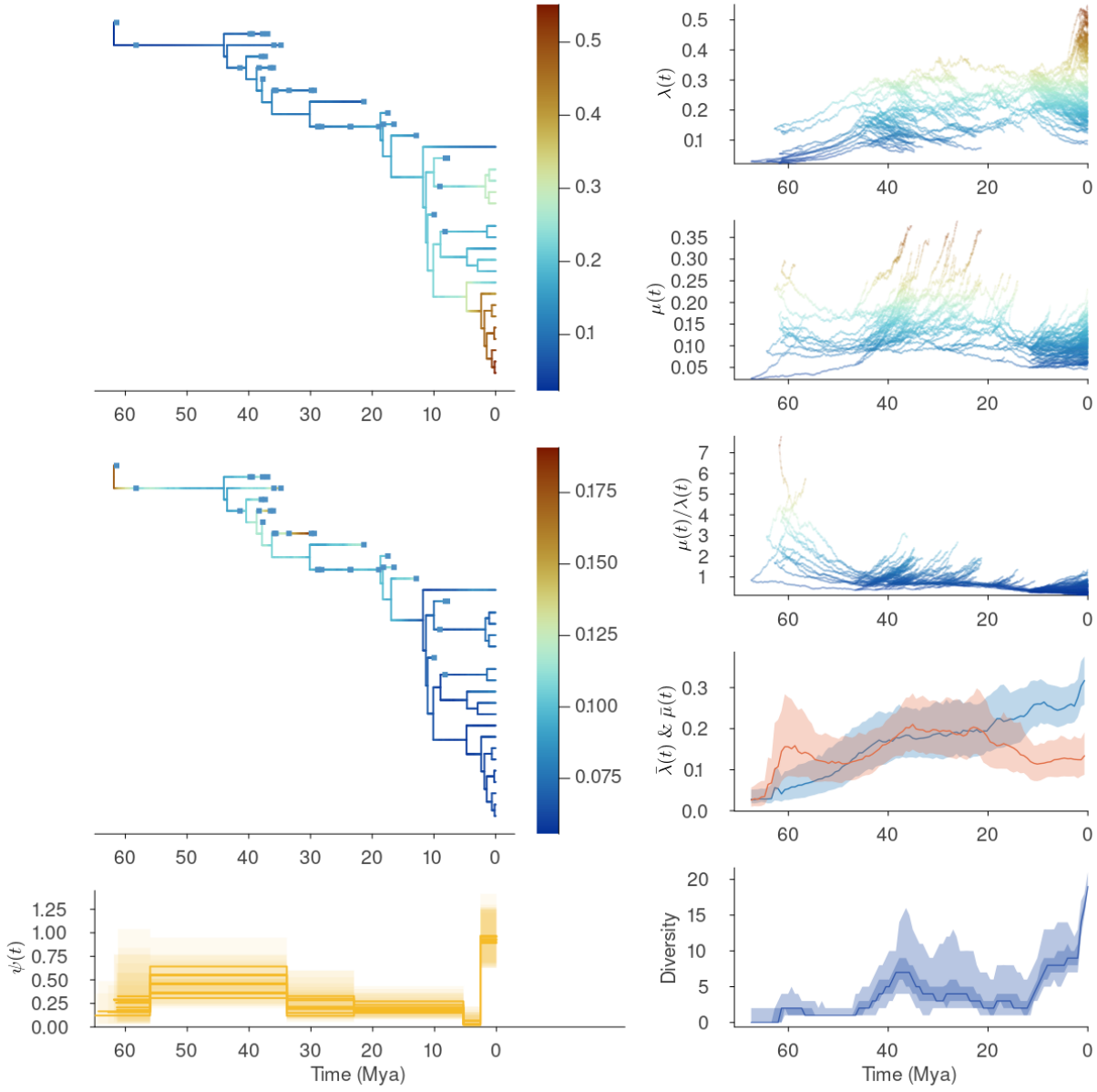
Supplementary Figure 13: FBDD results for Mauritiinae. Details as in fig. S3.



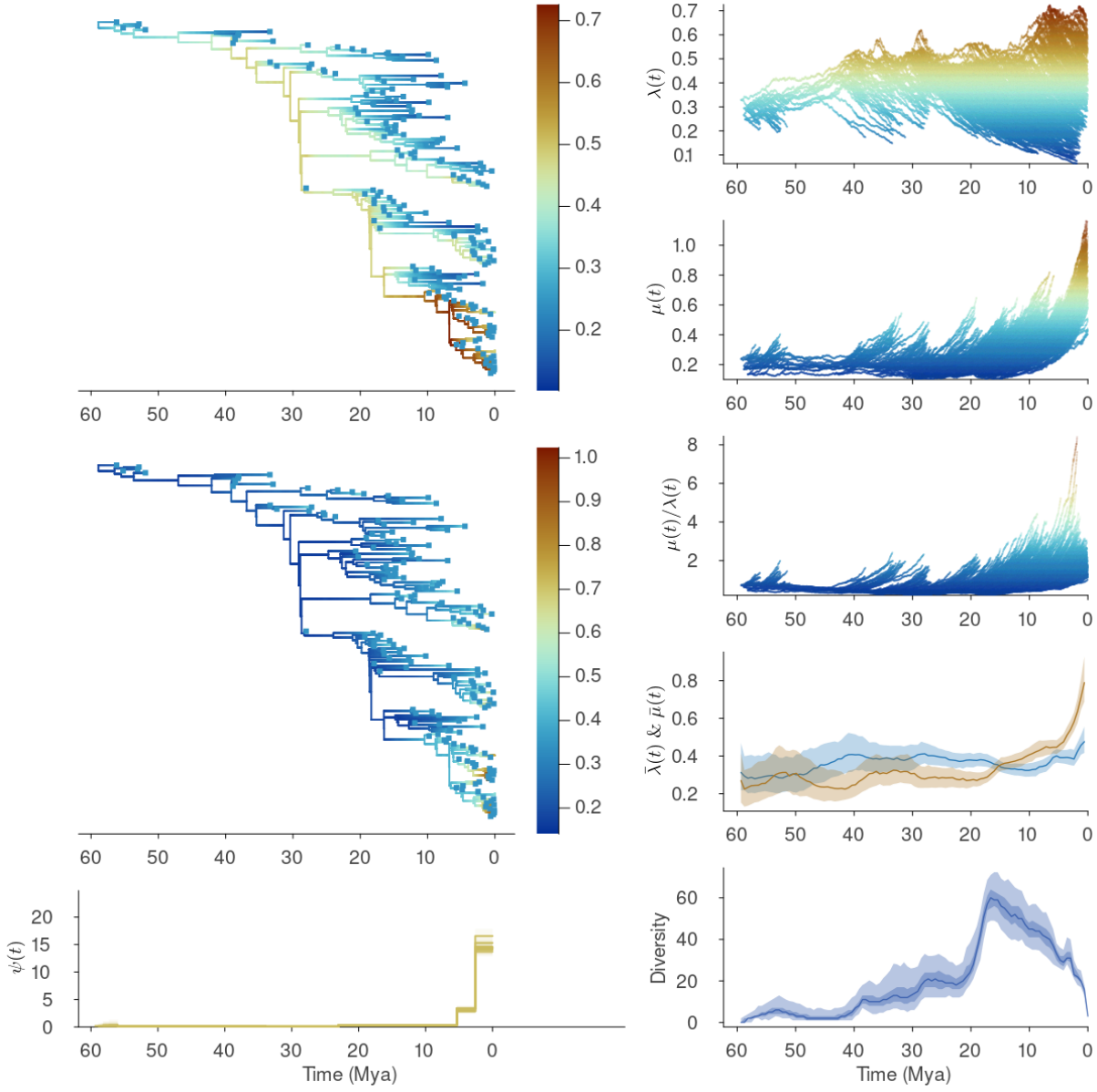
Supplementary Figure 14: FBDD results for Acanthuridae. Details as in fig. S3.



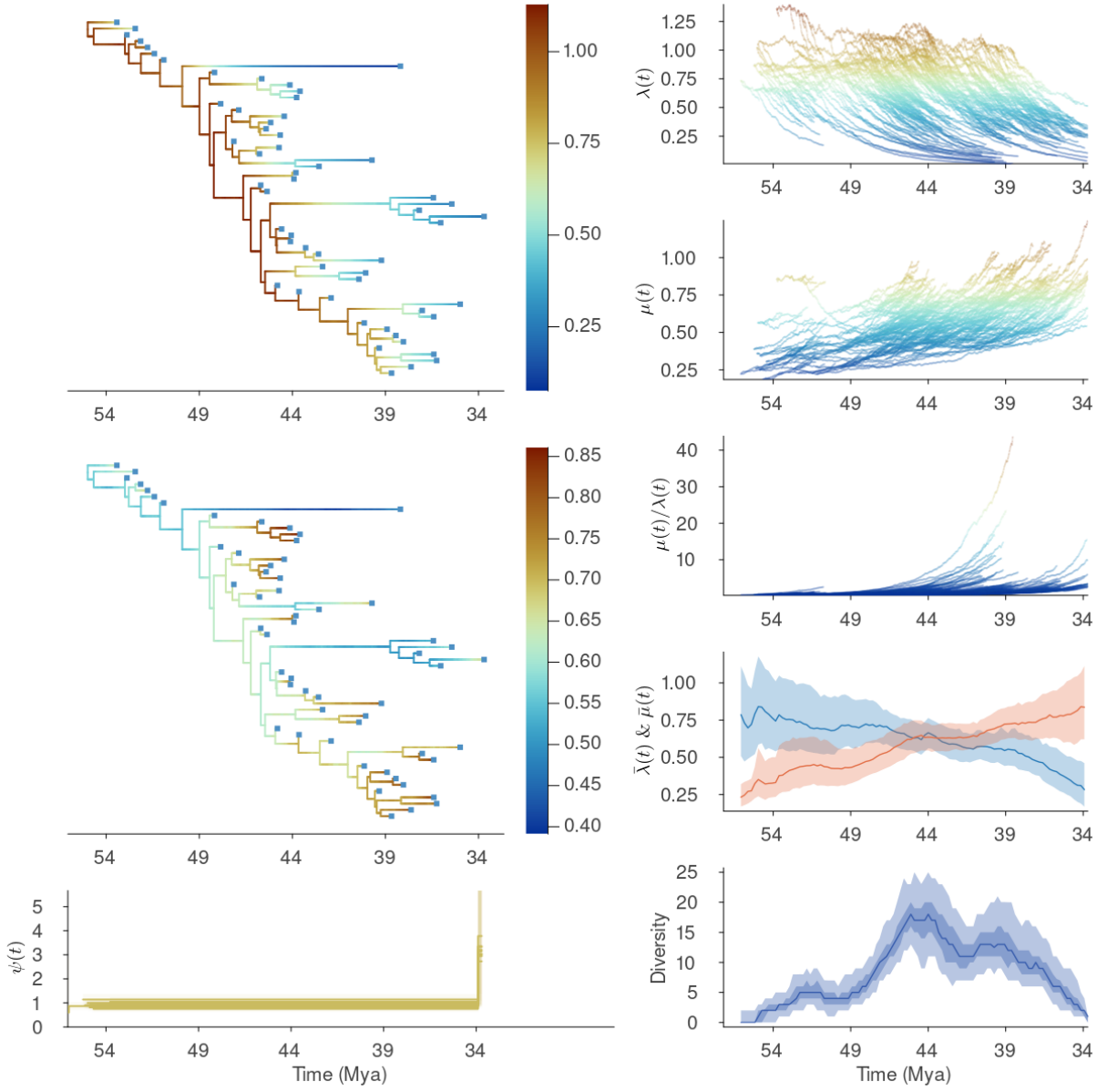
Supplementary Figure 15: FBDD results for Siganidae. Details as in fig. S3.



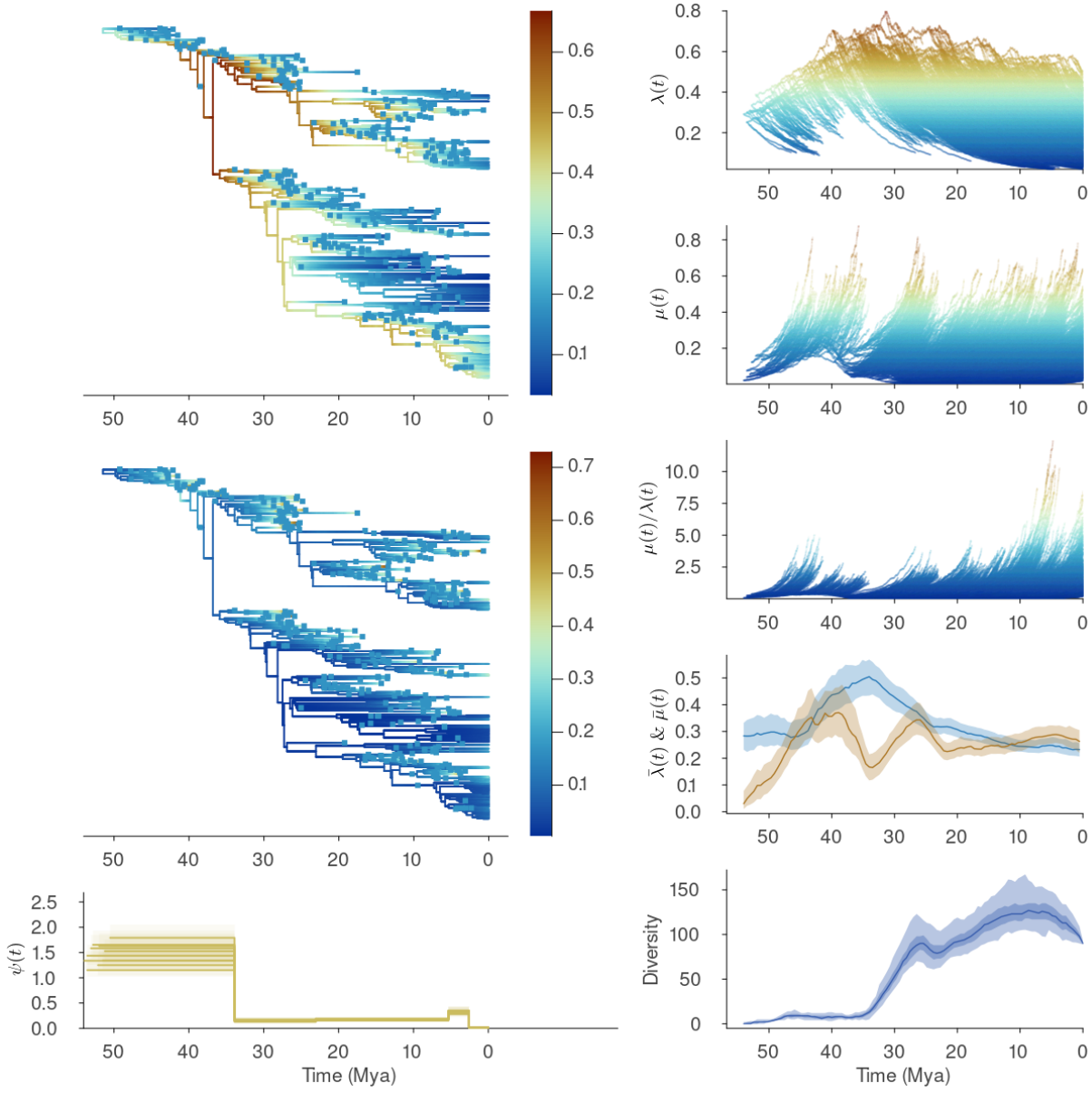
Supplementary Figure 16: FBDD results for Spheniscidae. Details as in fig. S3.



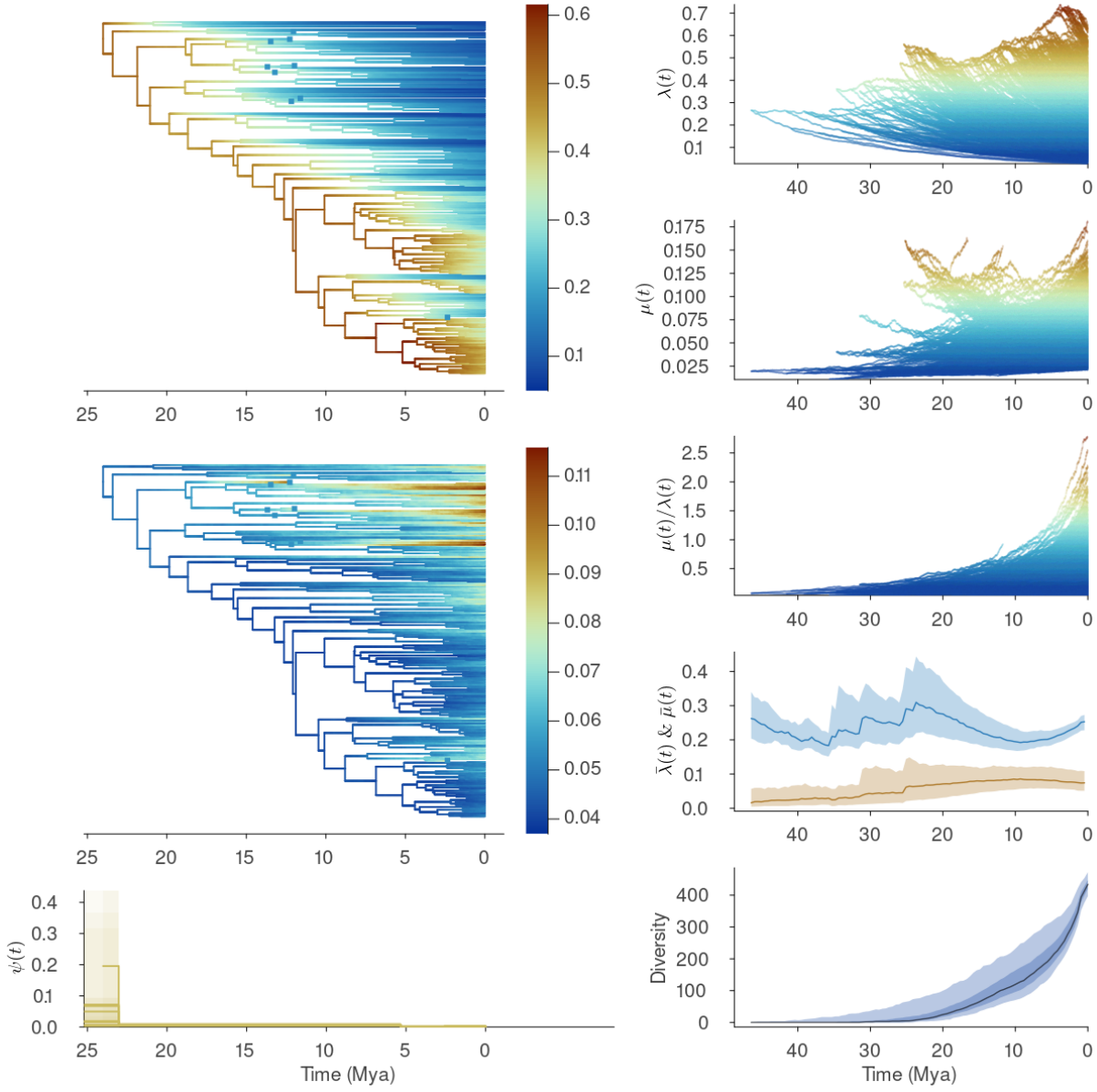
Supplementary Figure 17: FBDD results for Proboscidea. Details as in fig. S3.



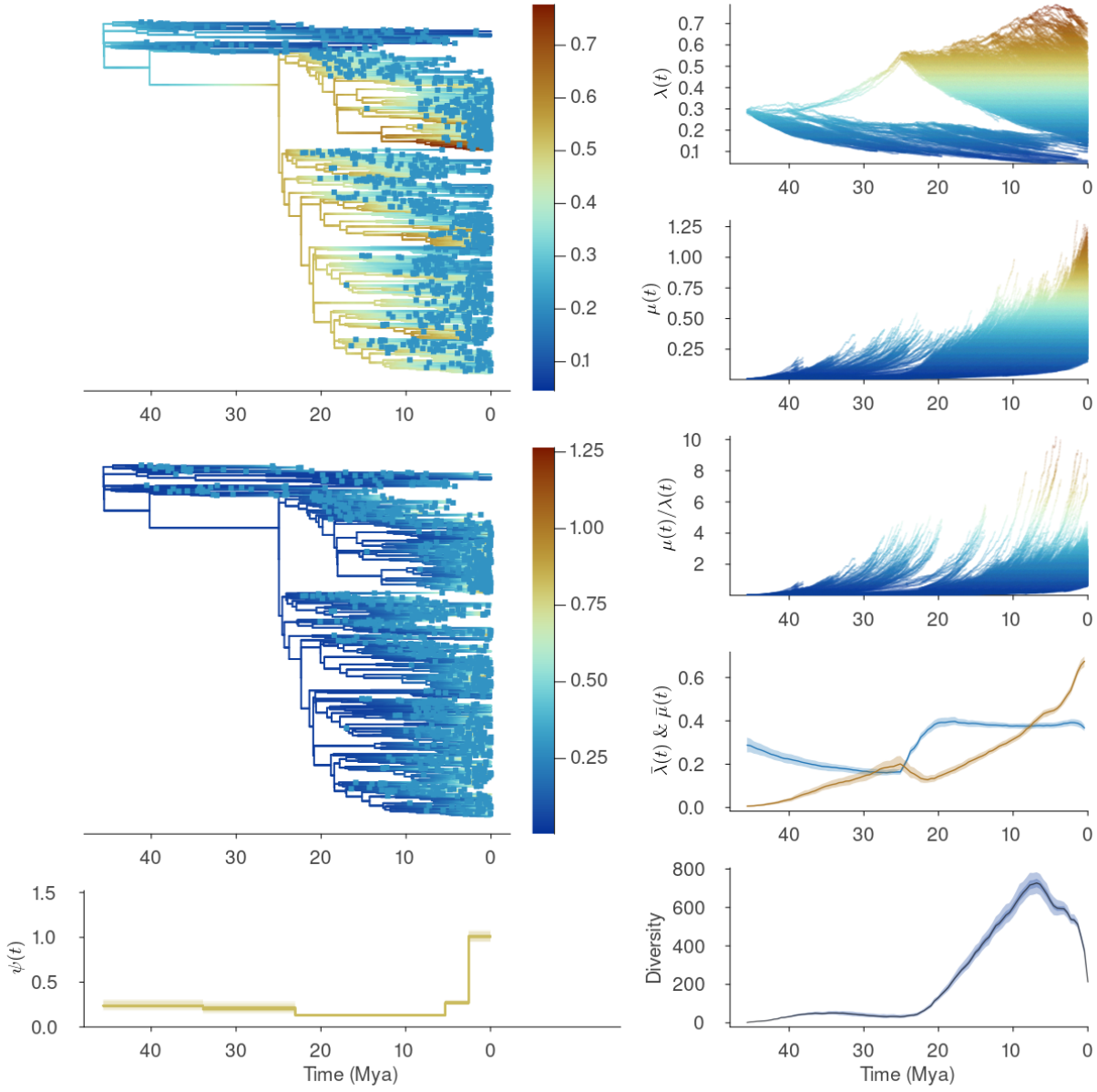
Supplementary Figure 18: FBDD results for Brontotheriioidea. Details as in fig. S3.



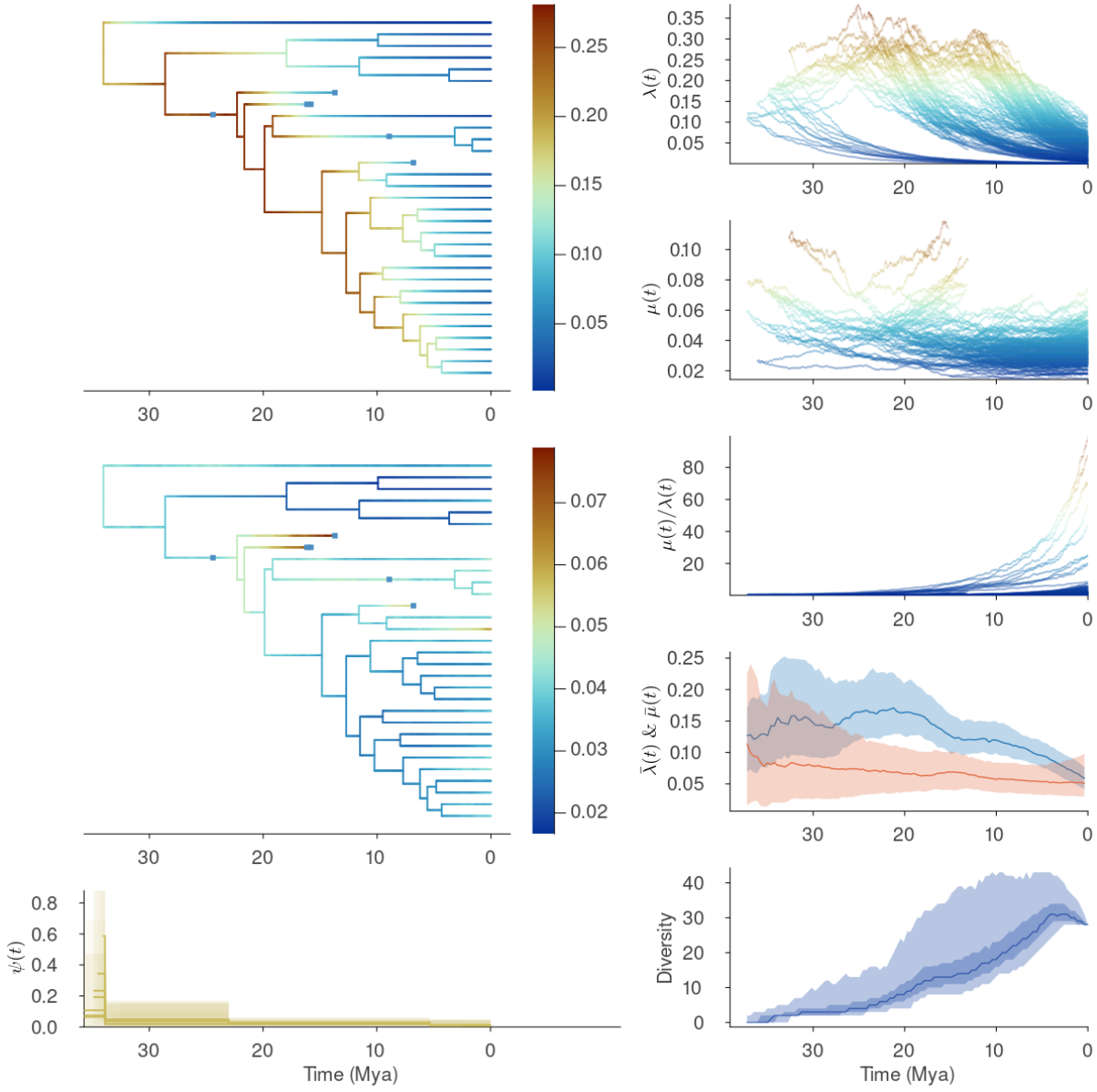
Supplementary Figure 19: FBDD results for Cetacea. Details as in fig. S3.



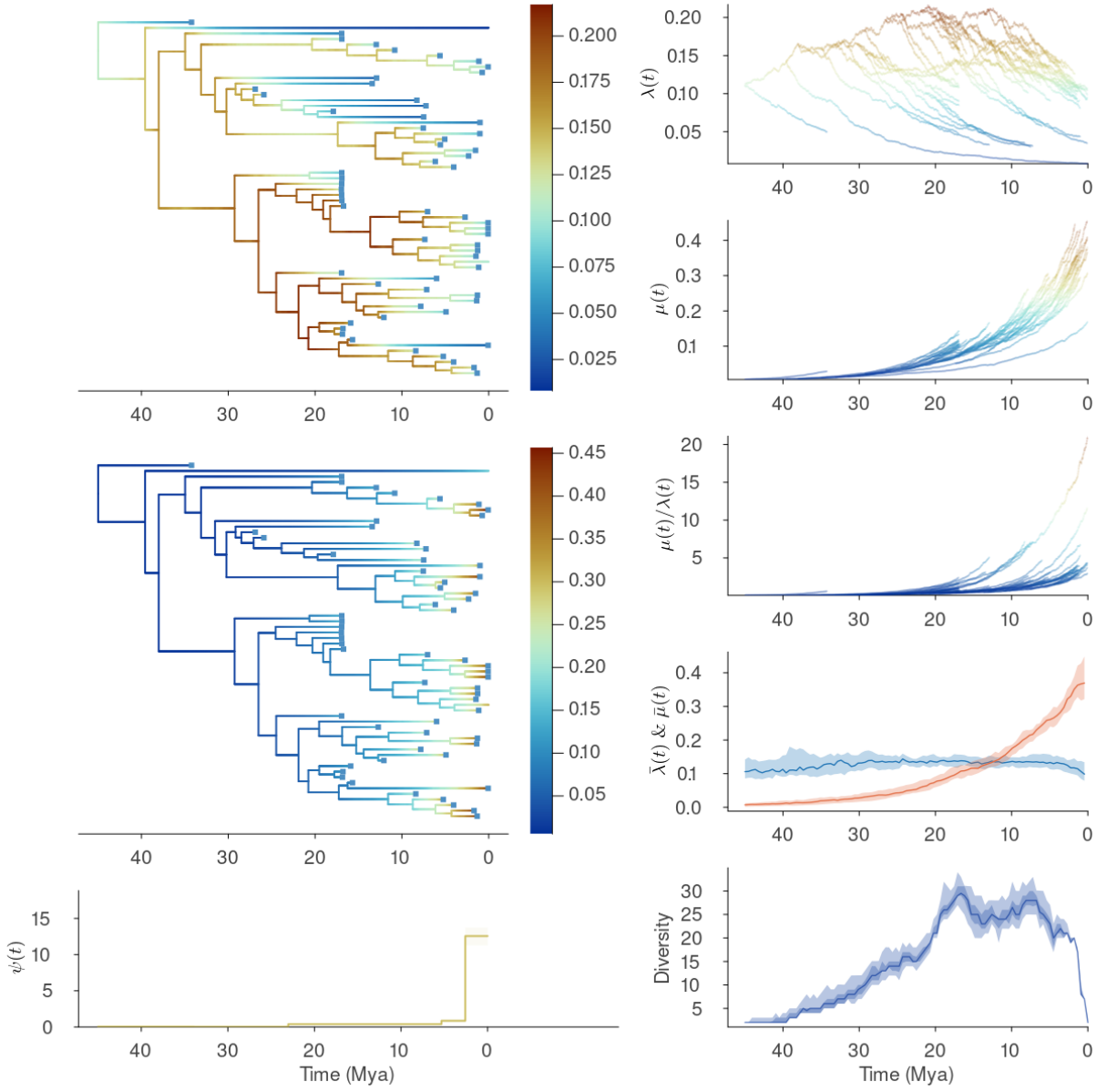
Supplementary Figure 20: FBDD results for Zoarcoidei. Details as in fig. S3.



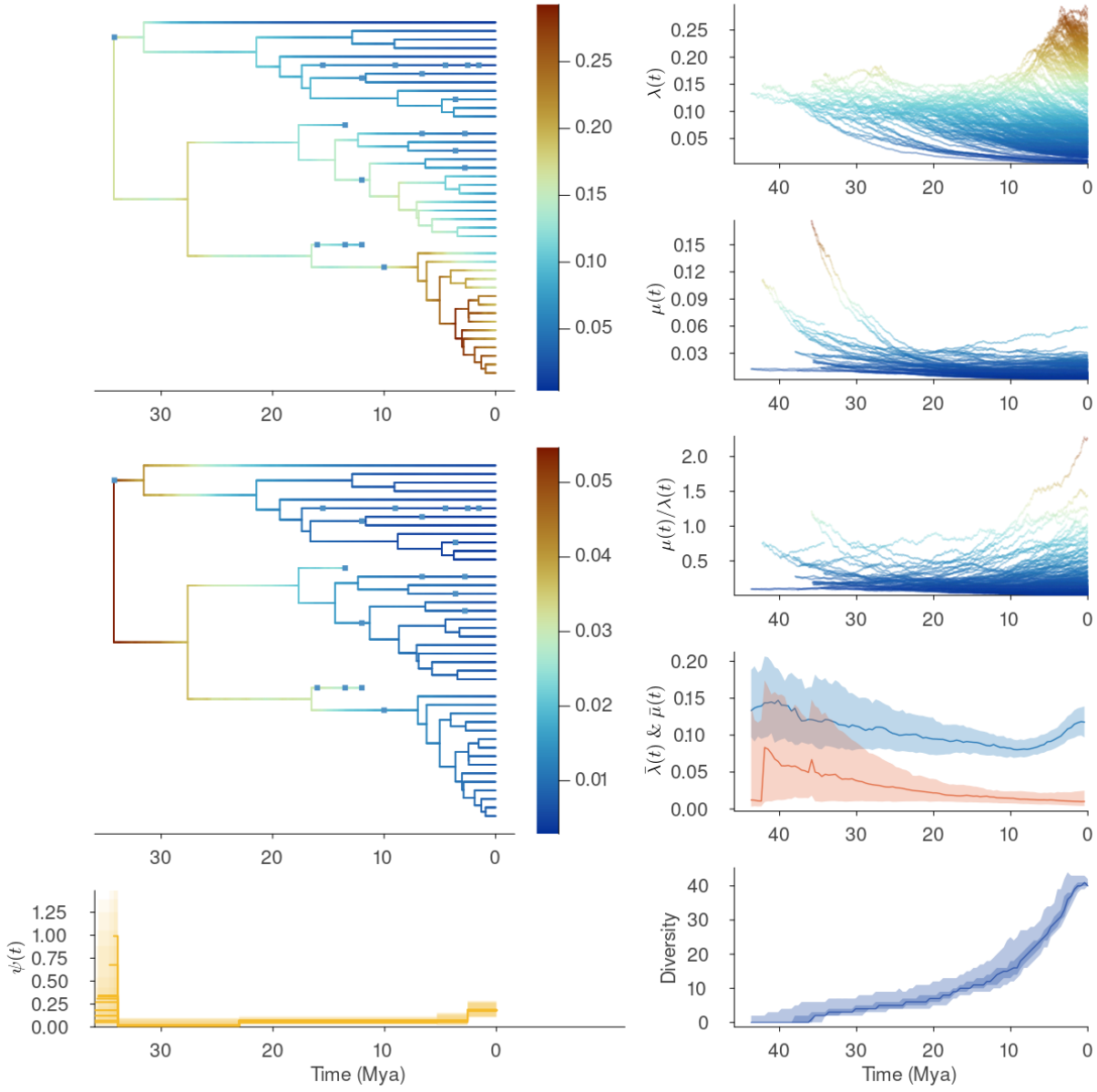
Supplementary Figure 21: FBDD results for Ruminantia. Details as in fig. S3.



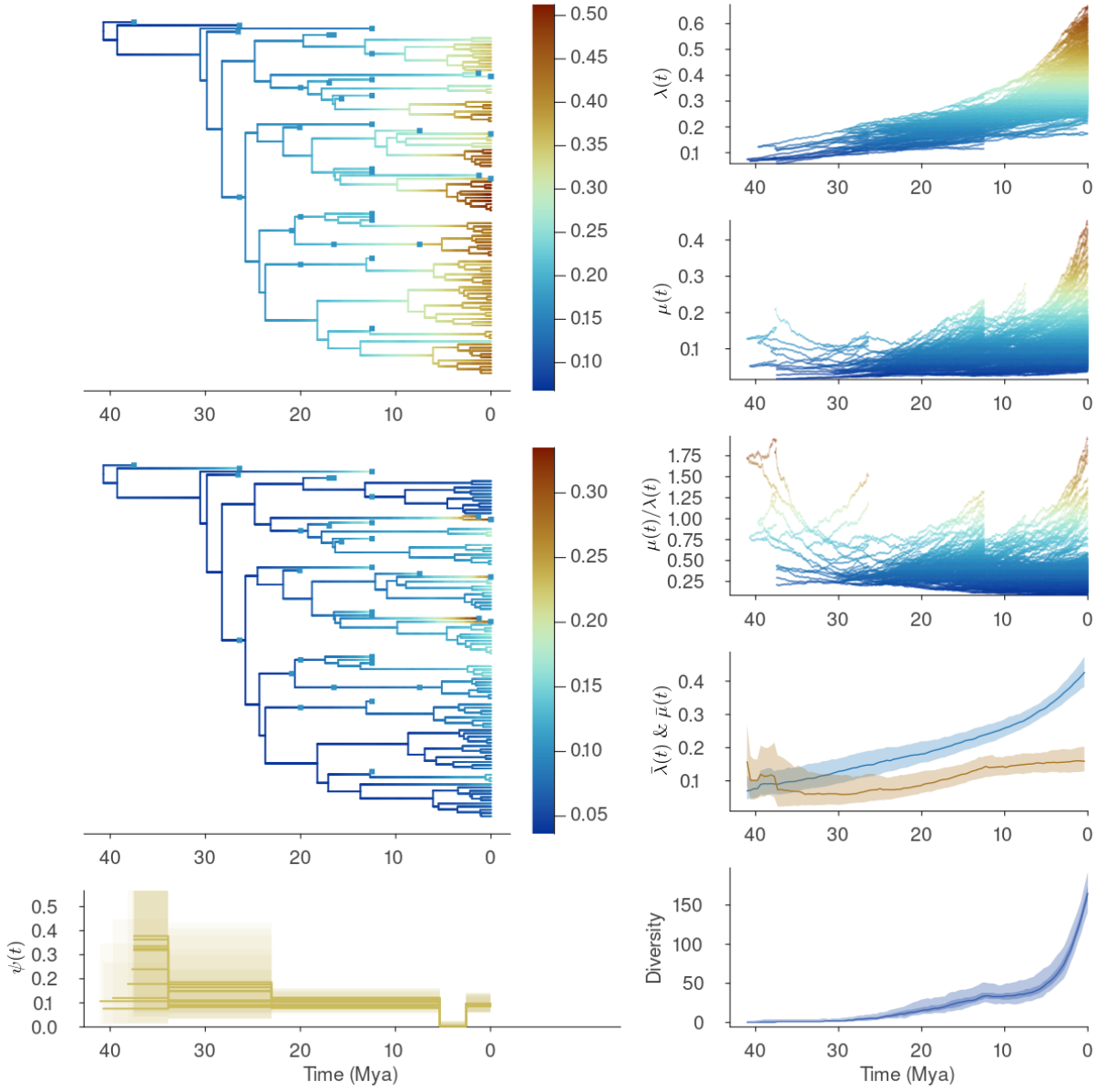
Supplementary Figure 22: FBDD results for Sthenurinae. Details as in fig. S3.



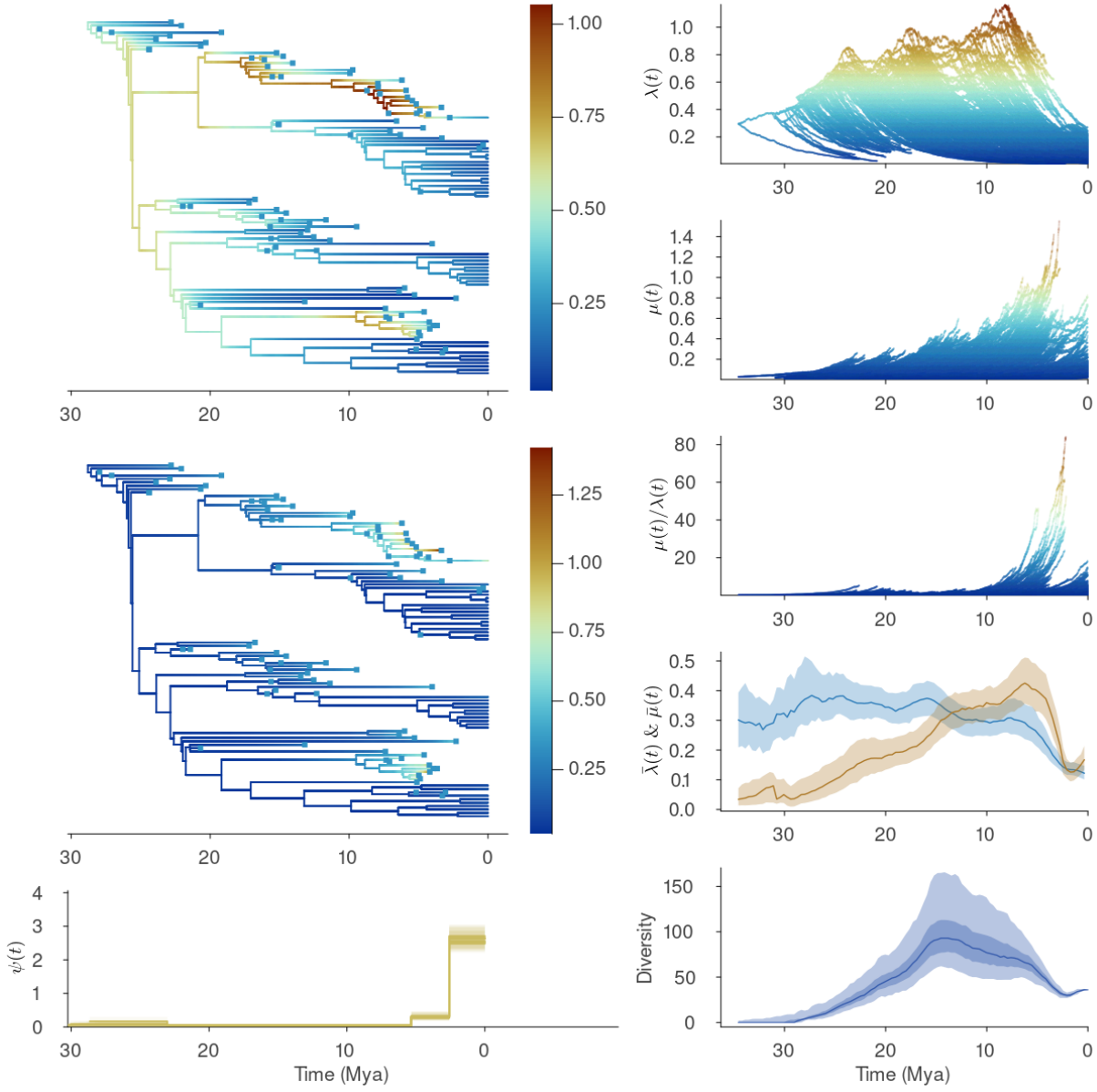
Supplementary Figure 23: FBDD results for Folivora. Details as in fig. S3.



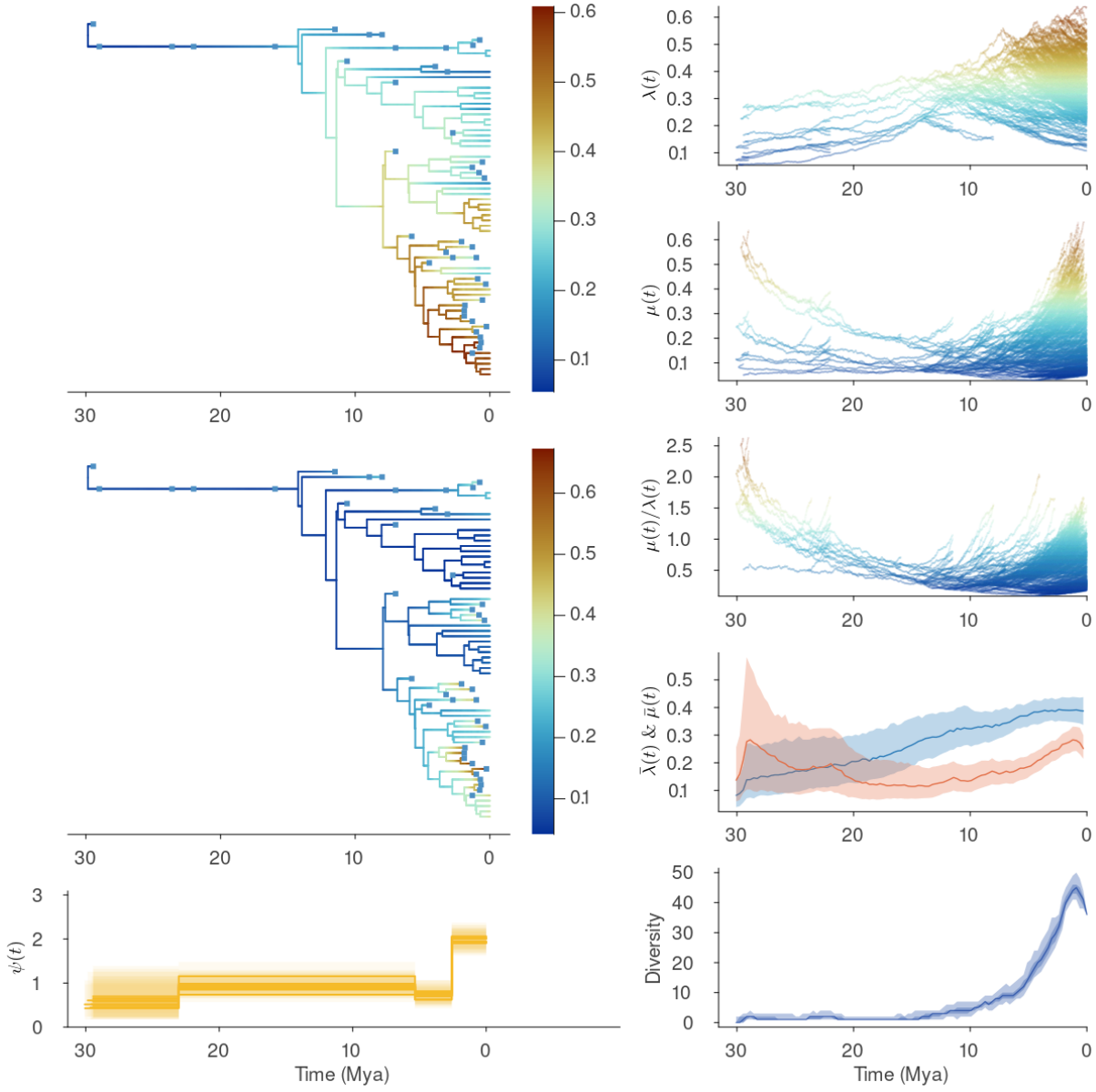
Supplementary Figure 24: FBDD results for Centrarchidae. Details as in fig. S3.



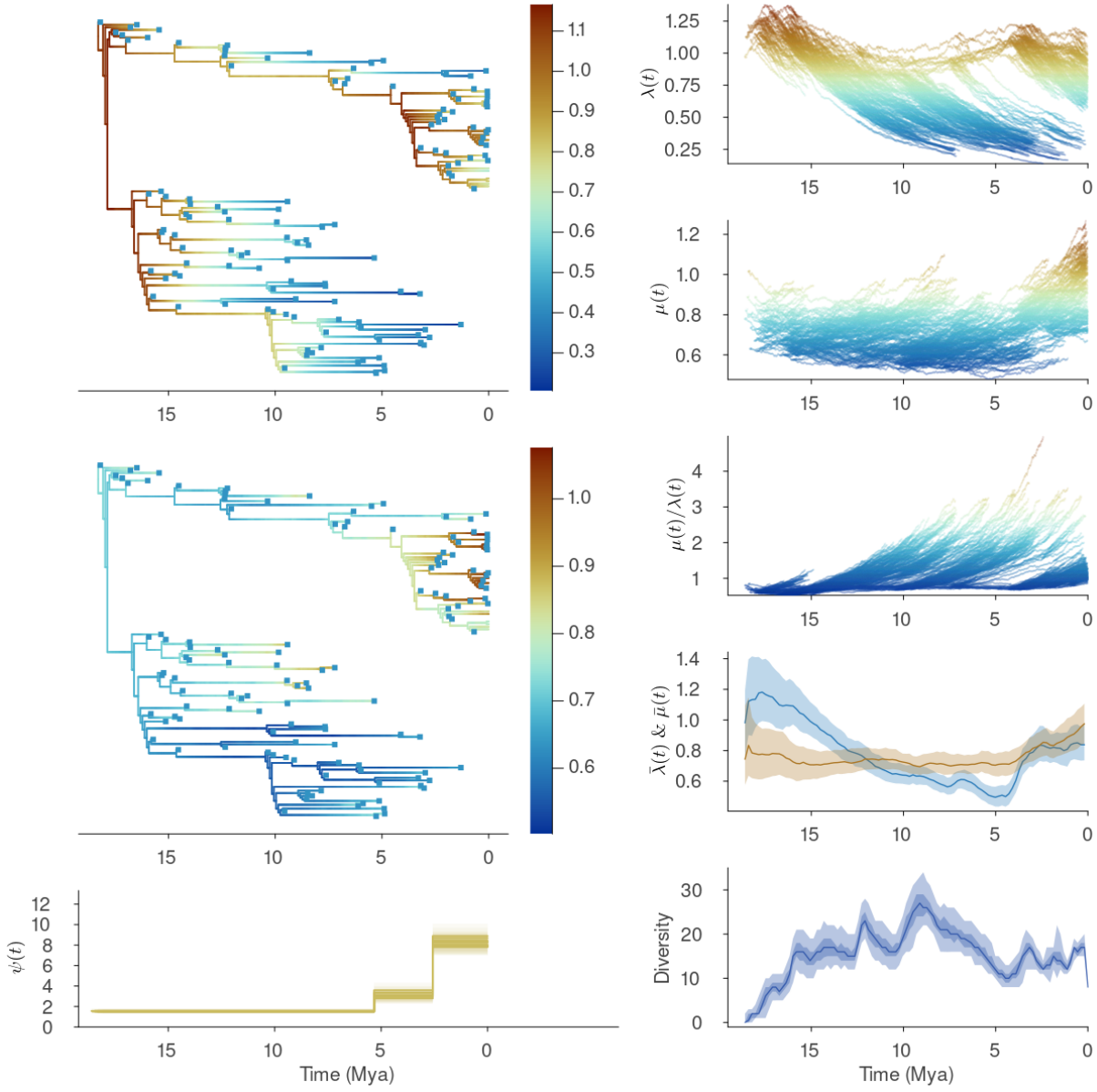
Supplementary Figure 25: FBDD results for Platyrrhines. Details as in fig. S3.



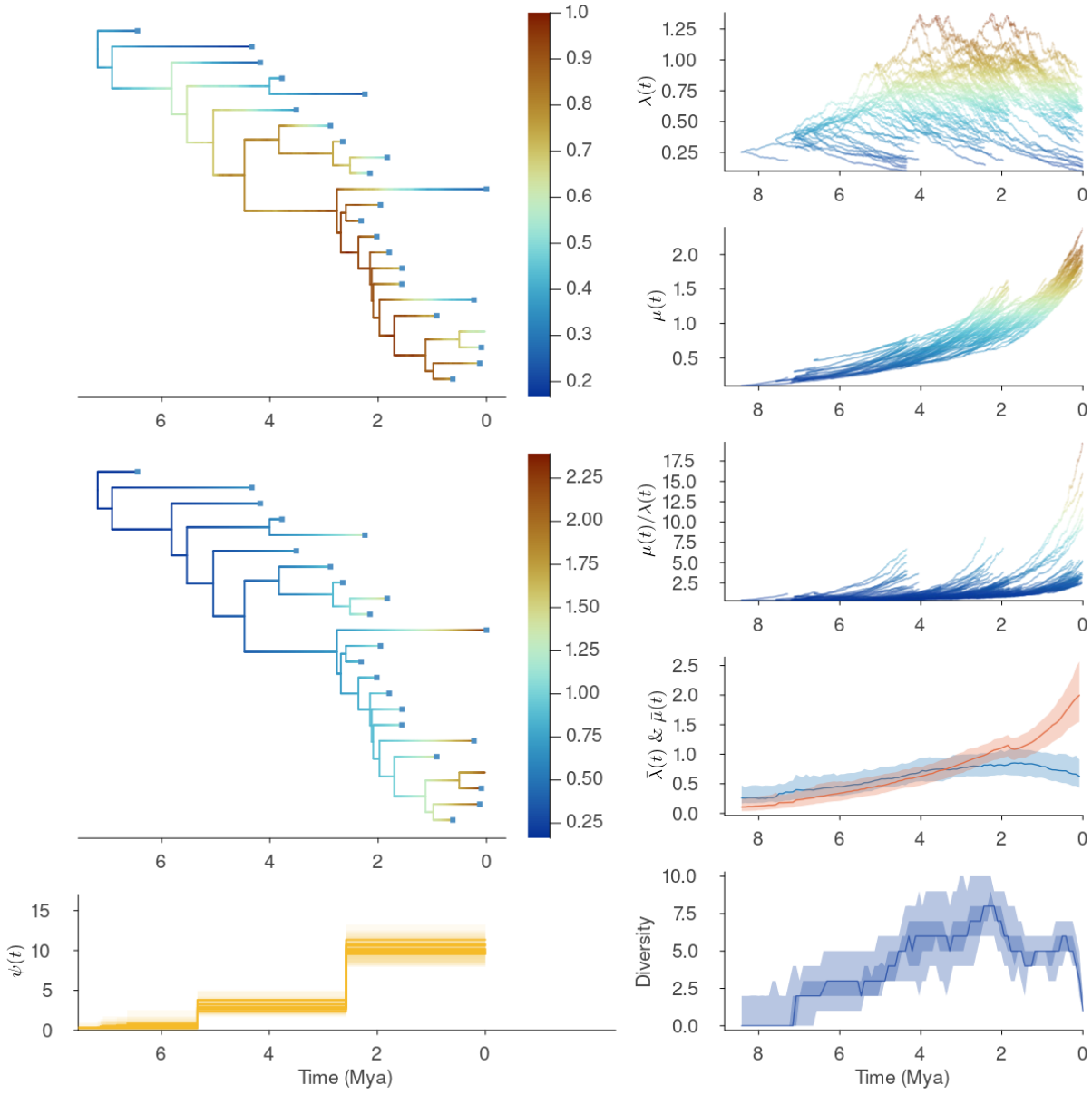
Supplementary Figure 26: FBDD results for Pinnipedia. Details as in fig. S3.



Supplementary Figure 27: FBDD results for Caninae. Details as in fig. S3.



Supplementary Figure 28: FBDD results for Equinae. Details as in fig. S3.



Supplementary Figure 29: FBDD results for Hominini. Details as in fig. S3.